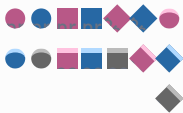


CENTER FOR FREE-ELECTRON LASER SCIENCE

GENERAL INFORMATION

Short label for the unit	en	CFEL
	de	CFEL
Official name of the organization	en	Center for Free-Electron Laser Science
	de	Center for Free-Electron Laser Science
One-line summary	en	Advances plasma and FEL accelerator science; hosts LUX and state-of-the-art laser labs.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Center for Free-Electron Laser Science		6	9	-	-	-	-	5	5	5	6	6	-	15
Group: Ultrafast Optics and X-rays		6	9	-	-	-	-	5	5	5	6	6	-	15
Team: Dielectric Laser Acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0
Team: Terahertz Acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

EUROPEAN ORGANIZATION FOR NUCLEAR RESEARCH

GENERAL INFORMATION

Short label for the unit	en	CERN
	de	CERN
Official name of the organization	en	European Organization for Nuclear Research
	de	Europäische Organisation für Kernforschung
One-line summary	en	Leads accelerator research; operates the Large Hadron Collider (LHC), SPS, and PS.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS										RECURSIVE SUM
European Organization for Nuclear Research	-	-	-	-	-	-	-	-	-	-	0
Sector: Accelerators and Technologies	-	-	-	-	-	-	-	-	-	-	0
Department: Beams	-	-	-	-	-	-	-	-	-	-	0
Group: Accelerators and Beam Physics	-	-	-	-	-	-	-	-	-	-	0
Section: Lepton Accelerators and Facilities	-	-	-	-	-	-	-	-	-	-	0
Section: Incoherent Effects	-	-	-	-	-	-	-	-	-	-	0
Section: Coherent Effects and Impedance	-	-	-	-	-	-	-	-	-	-	0
Section: Linear and Non-Linear Optics	-	-	-	-	-	-	-	-	-	-	0
Section: Non-Linear Dynamics and Collimation	-	-	-	-	-	-	-	-	-	-	0
Section: Hadron Sources and Linacs	-	-	-	-	-	-	-	-	-	-	0
Group: Administration, Safety and Resources	-	-	-	-	-	-	-	-	-	-	0
Section: Resources & Logistics (RL)	-	-	-	-	-	-	-	-	-	-	0
Section: Safety Unit	-	-	-	-	-	-	-	-	-	-	0
Group: Controls Software & Services	-	-	-	-	-	-	-	-	-	-	0
Section: Configuration, Settings & Automation	-	-	-	-	-	-	-	-	-	-	0
Section: Communication, Processing & Archival	-	-	-	-	-	-	-	-	-	-	0
Section: Front-end Software & Timing	-	-	-	-	-	-	-	-	-	-	0
Section: GUI Technologies & Applications	-	-	-	-	-	-	-	-	-	-	0
Section: Infrastructure & System Administration	-	-	-	-	-	-	-	-	-	-	0
Section: Software Engineering Technologies	-	-	-	-	-	-	-	-	-	-	0
Section: Data Science for Beam Operation	-	-	-	-	-	-	-	-	-	-	0
Group: Controls Electronics and Mechatronics	-	-	-	-	-	-	-	-	-	-	0
Section: Electronics Design & Low level software	-	-	-	-	-	-	-	-	-	-	0
Section: Electronics Production & Radiation Tolerant	-	-	-	-	-	-	-	-	-	-	0
Section: Infrastructure	-	-	-	-	-	-	-	-	-	-	0
Section: Mechatronics, Robotics & Operation	-	-	-	-	-	-	-	-	-	-	0
Section: Measurement, Test & Analysis	-	-	-	-	-	-	-	-	-	-	0

NAME	MEMBERS	SSOR	HER	DOC	JENT	JENT	JENT	ICIST	VEER	HER	MALE	MALE	MARY	SUM
Group: Operations		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Engineering		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Access & Alarm		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Accelerator Coordination and Engineering		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Cooling and Ventilation		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Electrical Engineering		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Handling Engineering		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Planning Administration & Safety		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Information Management		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Mechanical and Materials Engineering		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: LHC Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: ISOLDE Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: AD & ELENA Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: PS Operation & AD/ELENA Backup Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: LINAC4 & PSB Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: SPS & LEIR Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Technical Infrastructure Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Geodetic Metrology		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Acquisition, Processing and data Control Software		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Accelerator Surveying and Geodetic measurements		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Experiment Surveying and Alignment		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: High Precision Alignment technologies		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Experimental Areas		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Industrial Control Systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Automatic Control and Safety		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Engineering of Control Systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: SCADA Applications and Service management		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: SCADA Technologies and Frameworks		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Tools and Middleware for Automation		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Accelerator Systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Beam Instrumentation		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Beam Transfer		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Radio Frequency		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Power Converters		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Sources, Targets & Interactions		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Technology		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Cryogenics		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Machine Protection & Electrical Integrity		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Magnets, Superconductors & Cryostats		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Vacuum, Surfaces & Coating		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. CERN Accelerator Complex

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	CERN Accelerator Complex
	de	CERN Accelerator Complex
tagline	en	CERN Accelerator Complex
	de	CERN-Beschleunigerkomplex
description	en	
	de	

02. AD

Type	en	Particle Decelerator
	de	Teilchenverlangsamer
current life cycle	en	3.1 Operation
label	en	AD
	de	AD
tagline	en	Antiproton Decelerator
	de	Antiprotonen-Abbremsanlage
description	en	The Antiproton Decelerator at CERN is a facility used to produce low-energy antiprotons for antimatter research. It takes high-energy antiprotons from proton collisions and slows them down from just below the speed of light to about one-tenth the speed of light. This deceleration is achieved by using strong electric fields. In addition, beam cooling techniques are employed to reduce the spread in energy and deviation from the desired track.
	de	Der Antiprotonen-Entschleuniger am CERN ist eine Anlage zur Erzeugung von niederenergetischen Antiprotonen für die Antimaterieforschung. Er verwendet hochenergetische Antiprotonen aus Protonenkollisionen und bremst sie von knapp unter Lichtgeschwindigkeit auf etwa ein Zehntel der Lichtgeschwindigkeit ab. Diese Abbremsung wird durch den Einsatz starker elektrischer Felder erreicht. Darüber hinaus werden Techniken zur Strahlkühlung eingesetzt, um die Energiespreizung und die Abweichung von der gewünschten Bahn zu verringern.

03. ELENA

Type	en	Particle Decelerator
	de	Teilchenverlangsamer
current life cycle	en	3.1 Operation
label	en	ELENA
	de	ELENA
tagline	en	Extra Low Energy Antiproton ring
	de	Extra-Niedrigenergie-Antiprotonen-Ring
description	en	ELENA (Extra Low ENergy Antiproton) is a 30 m hexagonal storage ring that decelerates antiproton beams and delivers it to different experiments. It is designed to further decelerate the antiproton beam coming from the Antiproton Decelerator AD to an energy of 0.1 MeV for more precise measurements.
	de	ELENA (Extra Low ENergy Antiproton) ist ein 30 m langer sechseckiger Speicherring, der Antiprotonenstrahlen abbremst und an verschiedene Experimente liefert. Er soll den vom Antiproton Decelerator AD kommenden Antiprotonenstrahl für präzisere Messungen auf eine Energie von 0,1 MeV abbremsen.

04. ISOLDE

Type	en	Particle Decelerator
	de	Teilchenverlangsamer
current life cycle	en	3.1 Operation
label	en	ISOLDE
	de	ISOLDE
tagline	en	Isotope Separator On Line DEvice
	de	Einrichtung zur Erzeugung radioaktiver Ionenstrahlen
description	en	Facility for the production of radioactive nuclides by spallation, fission or fragmentation reactions in a thick target irradiated with a proton beam with an energy of 1.4 GeV and an intensity of up to 2 microamperes.
	de	Anlage zur Erzeugung radioaktiver Nuklide durch Spallations-, Spaltungs- oder Fragmentierungsreaktionen in einem dicken Target, das mit einem Protonenstrahl mit einer Energie von 1,4 GeV und einer Intensität bis zu 2 Mikroampere bestrahlt wird.

05. LHC

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	3.1 Operation
label	en	LHC
	de	LHC
tagline	en	Large Hadron Collider (LHC)
	de	Großer Hadronen-Collider (LHC)
description	en	
	de	

06. LHC Run 1

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	4.1 Upgrade
label	en	LHC Run 1
	de	LHC Run 1
tagline	en	Large Hadron Collider (LHC Run 1)
	de	Großer Hadronen-Collider
description	en	
	de	

07. LHC Run 2

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	4.1 Upgrade
label	en	LHC Run 2
	de	LHC Run 2
tagline	en	Large Hadron Collider (LHC Run 2)
	de	Großer Hadronen-Collider
description	en	
	de	

08. LHC Run 3

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	3. Operation And Maintenance
label	en	LHC Run 3
	de	LHC Run 3
tagline	en	Large Hadron Collider (LHC Run 3)
	de	Großer Hadronen-Collider
description	en	
	de	

09. HL-LHC (LHC Run 5)

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	1.2 Technical Design
label	en	HL-LHC (LHC Run 5)
	de	HL-LHC (LHC Run 5)
tagline	en	High-Luminosity Large Hadron Collider (LHC Run 5)
	de	Hochluminöser großer Hadronen-Collider
description	en	Der HL-LHC ist eine genehmigte Luminositätserhöhung des LHC. Der HL-LHC wird der erste Beschleuniger mit Quadrupol- und Dipolmagneten auf Basis von Nb3Sn-Supraleitern und damit ein wichtiges technologisches Sprungbrett für längerfristige Zukunftsprojekte wie FCC-hh und HE-LHC darstellen. Der HL-LHC wird auch der erste Hadronenbeschleuniger sein, der Krabbenhohlräume verwendet.
	de	Der HL-LHC ist eine genehmigte Luminositätserhöhung des LHC. Der HL-LHC wird der erste Beschleuniger mit Quadrupol- und Dipolmagneten auf Basis von Nb3Sn-Supraleitern und damit ein wichtiges technologisches Sprungbrett für längerfristige Zukunftsprojekte wie FCC-hh und HE-LHC darstellen. Der HL-LHC wird auch der erste Hadronenbeschleuniger sein, der Krabbenhohlräume verwendet.

10. HL-LHC (LHC Run 4)

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	2.1 Preparation
label	en	HL-LHC (LHC Run 4)
	de	HL-LHC (LHC Run 4)
tagline	en	High-Luminosity Large Hadron Collider (LHC Run 4)
	de	Hochluminöser großer Hadronen-Collider
description	en	The HL-LHC is an approved luminosity increase of the LHC. The HL-LHC will be the first accelerator with quadrupole and dipole magnets based on Nb3Sn superconductors and thus an important technological stepping stone for longer-term future projects such as FCC-hh and HE-LHC. The HL-LHC will also be the first hadron collider to use crab cavities.
	de	Der HL-LHC ist eine genehmigte Luminositätserhöhung des LHC. Der HL-LHC wird der erste Beschleuniger mit Quadrupol- und Dipolmagneten auf Basis von Nb3Sn-Supraleitern und damit ein wichtiges technologisches Sprungbrett für längerfristige Zukunftsprojekte wie FCC-hh und HE-LHC darstellen. Der HL-LHC wird auch der erste Hadronenbeschleuniger sein, der Krabbenhohlräume verwendet.

11. Linac 1

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	Linac 1
	de	Linac 1
tagline	en	Linear accelerator 1
	de	Linearbeschleuniger 1
description	en	
	de	

12. Linac 2

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	Linac 2
	de	Linac 2
tagline	en	Linear accelerator 2
	de	Linearbeschleuniger 2
description	en	
	de	

13. Linac 3

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	Linac 3
	de	Linac 3
tagline	en	Linear accelerator 3
	de	Linearbeschleuniger 3
description	en	
	de	

14. Linac 4

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	Linac 4
	de	Linac 4
tagline	en	Linear accelerator 4
	de	Linearbeschleuniger 4
description	en	
	de	

15. PS

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	PS
	de	PS
tagline	en	Proton Synchrotron
	de	Protonen-Synchrotron
description	en	
	de	

16. PSB

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	PSB
	de	PSB
tagline	en	Proton Synchrotron Booster
	de	Proton Synchrotron Booster
description	en	
	de	

17. SPS

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	SPS
	de	SPS
tagline	en	Super Proton Synchrotron
	de	Super-Protonen-Synchrotron
description	en	
	de	

18. CLIC

Type	en	
	de	
current life cycle	en	1.2.2 Technically Designed
label	en	CLIC
	de	CLIC
tagline	en	Compact Linear Collider
	de	Kompakter Linearcollider
description	en	Linearer Elektronen-Positronen-Collider im TeV-Bereich mit hoher Luminosität für die Erforschung kleinster Teilchen, der von internationalen Kollaborationen unter der Leitung des CERN entwickelt wird. Der CLIC-Beschleuniger wurde für drei Energiestufen bei Massenschwerpunktenergien von 380 GeV, 1,5 TeV und 3 TeV optimiert [1]. CLIC verwendet eine neuartige Zweistrahl-Beschleunigungstechnik mit normaleitenden Beschleunigungsstrukturen, die im Bereich von 70 MV/m bis 100 MV/m arbeiten.
	de	Linearer Elektronen-Positronen-Collider im TeV-Bereich mit hoher Luminosität für die Erforschung kleinster Teilchen, der von internationalen Kollaborationen unter der Leitung des CERN entwickelt wird. Der CLIC-Beschleuniger wurde für drei Energiestufen bei Massenschwerpunktenergien von 380 GeV, 1,5 TeV und 3 TeV optimiert [1]. CLIC verwendet eine neuartige Zweistrahl-Beschleunigungstechnik mit normaleitenden Beschleunigungsstrukturen, die im Bereich von 70 MV/m bis 100 MV/m arbeiten.

19. FCC

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	1.2.1 Technical Design Creation
label	en	FCC
	de	FCC
tagline	en	Future Circular Collider
	de	Zukünftiger Ringkollider
description	en	Future Circular Collider (ee)
	de	Future Circular Collider (ee)

20. FCC-ee

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	1.2.1 Technical Design Creation
label	en	FCC-ee
	de	FCC-ee
tagline	en	Future Circular Collider (ee)
	de	Zukünftiger Elektronen-Positronen-Ringkollider
description	en	Future Circular Collider (ee)
	de	Future Circular Collider (ee)

21. FCC-hh

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	1.2.1 Technical Design Creation
label	en	FCC-hh
	de	FCC-hh
tagline	en	Future Circular Collider (hh)
	de	Zukünftiger Hadronen-Ringkollider
description	en	Future Circular Collider (hh)
	de	Future Circular Collider (hh)

TEST FACILITIES

01. AWAKE

Type	en	Plasma-Based Beam-Driven Acceleration
	de	plasmabasierte teilchenstrahlgetriebene Beschleunigung
current life cycle	en	4.1 Upgrade
label	en	AWAKE
	de	AWAKE
tagline	en	Advanced proton driven plasma Wakefield Acceleration Experiment
	de	Fortgeschrittenes plasmabasiertes protonengetriebenes Beschleunigungsexperiment
description	en	Feasibility study on the use of plasma wakefields driven by a proton beam to accelerate charged particles.
	de	Machbarkeitsstudie über die Verwendung von Plasma-Wakefields, die von einem Protonenstrahl angetrieben werden, um geladene Teilchen zu beschleunigen.

02. AWAKE (Run 1)

Type	en	Plasma-Based Beam-Driven Acceleration
	de	plasmabasierte teilchenstrahlgetriebene Beschleunigung
current life cycle	en	4.1 Upgrade
label	en	AWAKE (Run 1)
	de	AWAKE (Run 1)
tagline	en	Advanced proton driven plasma Wakefield Acceleration Experiment
	de	Fortgeschrittenes plasmabasiertes protonengetriebenes Beschleunigungsexperiment
description	en	Feasibility study on the use of plasma wakefields driven by a proton beam to accelerate charged particles.
	de	Machbarkeitsstudie über die Verwendung von Plasma-Wakefields, die von einem Protonenstrahl angetrieben werden, um geladene Teilchen zu beschleunigen.

03. AWAKE

Type	en	Plasma-Based Beam-Driven Acceleration
	de	plasmabasierte teilchenstrahlgetriebene Beschleunigung
current life cycle	en	3.1 Operation
label	en	AWAKE
	de	AWAKE (Run 2)
tagline	en	Advanced proton driven plasma Wakefield Acceleration Experiment
	de	Fortgeschrittenes plasmabasiertes protonengetriebenes Beschleunigungsexperiment
description	en	Feasibility study on the use of plasma wakefields driven by a proton beam to accelerate charged particles.
	de	Machbarkeitsstudie über die Verwendung von Plasma-Wakefields, die von einem Protonenstrahl angetrieben werden, um geladene Teilchen zu beschleunigen.

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

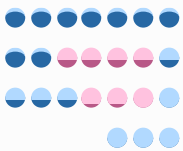
Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

DEUTSCHES ELEKTRONEN-SYNCHROTRON DESY

GENERAL INFORMATION

Short label for the unit	en	DESY
	de	DESY
Official name of the organization	en	Deutsches Elektronen-Synchrotron DESY
	de	Deutsches Elektronen-Synchrotron DESY
One-line summary	en	Pioneers high-intensity accelerators; operates PETRA III, FLASH, SINBAD, and hosts European XFEL.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Deutsches Elektronen-Synchrotron DESY		-	9	-	8	2	5	24	-	-	7	17	-	24
Division: Accelerators		-	9	-	8	2	5	24	-	-	7	17	-	24
Accelerator Operation Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Diagnostics and Instrumentation Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Control Systems Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Machine and experiment set-up		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: FLASH		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Cryogenics, superconductivity		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Injection		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Power, Cooling, Climate		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Lasers for plasma accelerators		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: High-frequency technology		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Power converters		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: PETRA III		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Plasma accelerators		-	9	-	8	2	5	24	-	-	7	17	-	24
Group: Beam controls		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Personal safety systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Superconducting accelerator technology		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Accelerator physics		-	-	-	-	-	-	-	-	-	-	-	-	0
XFEL		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Vacuum systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Accelerator Physics Zeuthen		-	-	-	-	-	-	-	-	-	-	-	-	0
PITZ		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. PETRA

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	4.1.2 Replaced By Upgrade
label	en	PETRA
	de	PETRA
tagline	en	Positron-Electron Tandem Ring Accelerator
	de	Positron-Electron Tandem Ringanlage
description	en	
	de	

02. PETRA III

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	3.1 Operation
label	en	PETRA III
	de	PETRA III
tagline	en	
	de	Synchrotronstrahlungsquelle der 3. Generation für harte Röntgenstrahlung
description	en	
	de	

03. PETRA IV

Type	en	Synchrotron Radiation Source (4th Generation)
	de	Synchrotronlichtquelle (4. Generation)
current life cycle	en	2.1 Preparation
label	en	PETRA IV
	de	PETRA IV
tagline	en	
	de	Synchrotronstrahlungsquelle der 4. Generation für harte Röntgenstrahlung
description	en	
	de	

04. REGAE

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	REGAE
	de	REGAE
tagline	en	Relativistic Electron Gun for Atomic Exploration
	de	Relativistische Elektronenquelle für die Untersuchung von Atomen
description	en	Source of ultra-short electron pulses for accelerator research and time-resolved diffraction experiments on crystals, liquids and surfaces. The acronym REGAE stands for „Relativistic Electron Gun for Atomic Exploration“.
	de	Quelle für ultrakurze Elektronenpulse für die Beschleunigungsforschung sowie für Experimente zur zeitaufgelösten Beugung an Kristallen, Flüssigkeiten und Oberflächen. Das Akronym REGAE steht für „Relativistic Electron Gun for Atomic Exploration“.

05. FLASH

Type	en	
	de	
current life cycle	en	4.1 Upgrade
label	en	FLASH
	de	FLASH
tagline	en	Free electron-laser in Hamburg
	de	Freie-Elektronen-Laser in Hamburg (FLASH)
description	en	Free electron-laser in Hamburg
	de	Free electron-laser in Hamburg

06. FLASH

Type	en	
	de	
current life cycle	en	4.1 Upgrade
label	en	FLASH
	de	FLASH
tagline	en	Free electron-laser in Hamburg
	de	Freie-Elektronen-Laser in Hamburg (FLASH)
description	en	Free electron-laser in Hamburg
	de	Free electron-laser in Hamburg

07. FLASH 1+2

Type	en	
	de	
current life cycle	en	4.1 Upgrade
label	en	FLASH 1+2
	de	FLASH 1+2
tagline	en	Free electron-laser in Hamburg
	de	Freie-Elektronen-Laser in Hamburg (FLASH 1+2)
description	en	Free electron-laser in Hamburg
	de	Free electron-laser in Hamburg

08. FLASH 2020+

Type	en	
	de	
current life cycle	en	2.3 Commissioning
label	en	FLASH 2020+
	de	FLASH 2020+
tagline	en	Free electron-laser in Hamburg
	de	Freie-Elektronen-Laser in Hamburg (FLASH 2020+)
description	en	Free electron-laser in Hamburg
	de	Free electron-laser in Hamburg

TEST FACILITIES

01. SINBAD

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	SINBAD
	de	SINBAD
tagline	en	Short INnovative Bunches and Accelerators at DESY
	de	Kurze innovative Teilchenpakete und Beschleuniger bei DESY
description	en	
	de	

02. ARES

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	ARES
	de	ARES
tagline	en	Accelerator Research Experiment at SINBAD
	de	Beschleuniger-Forschungsexperiment bei SINBAD
description	en	Linearbeschleuniger zur Erzeugung ultrakurzer Elektronenpakete. Teil von SINBAD bei DESY. Das Ziel von ARES ist es, ultrakurze Elektronenpakete mit geringer Ladung (0,2 - 50 pC) und hoher Ankunftszeitstabilität (weniger als 10 fs) für verschiedene Anwendungen zu erzeugen, z.B. für die externe Injektion zur Beschleunigung von Laserplasma-wakefieldss [2]. Die Grundkonzeption des Beschleunigers sieht einen S-Band-Photoinjektor vor, der Elektronenpakete mit geringer Ladung durch Geschwindigkeits-Bündelung komprimiert und auf 100 MeV Energie beschleunigt.
	de	Linearbeschleuniger zur Erzeugung ultrakurzer Elektronenpakete. Teil von SINBAD bei DESY. Das Ziel von ARES ist es, ultrakurze Elektronenpakete mit geringer Ladung (0,2 - 50 pC) und hoher Ankunftszeitstabilität (weniger als 10 fs) für verschiedene Anwendungen zu erzeugen, z.B. für die externe Injektion zur Beschleunigung von Laserplasma-wakefieldss [2]. Die Grundkonzeption des Beschleunigers sieht einen S-Band-Photoinjektor vor, der Elektronenpakete mit geringer Ladung durch Geschwindigkeits-Bündelung komprimiert und auf 100 MeV Energie beschleunigt.

03. AXSIS

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	AXSIS
	de	AXSIS
tagline	en	Attosecond X-ray Science, Imaging and Spectroscopy
	de	Attosekunden-Röntgenforschung, Bildgebung und Spektroskopie
description	en	15 MeV-Elektronenbeschleuniger auf der Grundlage von THz-Wellen in dielektrischen Wellenleitern
	de	15 MeV-Elektronenbeschleuniger auf der Grundlage von THz-Wellen in dielektrischen Wellenleitern

04. DESY Accelerator Comidex

Type	en	Accelerator Complex
	de	Beschleunigerkomplex
current life cycle	en	3.1 Operation
label	en	DESY Accelerator Comidex
	de	DESY-Beschleunigerkomplex
tagline	en	
	de	
description	en	
	de	

05. DESY

Type	en	
	de	
current life cycle	en	4.1.1 Being Replaced By Upgrade
label	en	DESY
	de	DESY
tagline	en	
	de	
description	en	
	de	

06. DESY II

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	DESY II
	de	DESY II
tagline	en	Preaccelerator for PETRA III
	de	Vorbeschleuniger für PETRA III
description	en	
	de	

07. DESY-II Test Beam Facility

Type	en	Beamline
	de	Beamline
current life cycle	en	3.1 Operation
label	en	DESY-II Test Beam Facility
	de	DESY-II Teststrahl-Anlage
tagline	en	
	de	
description	en	Die DESY-II Test Beam Facility ist eine der wenigen Teststrahlanlagen weltweit, die Elektronen und Positronen im GeV-Bereich bereit stellt. Umfangreiche Strahldiagnostik und andere Instrumentierung stehen bereit.
	de	Eine der wenigen Teststrahlanlagen weltweit, die Elektronen und Positronen im GeV-Bereich bereitstellt. Umfangreiche Strahldiagnostik und andere Instrumentierung stehen bereit.

08. DORIS

Type	en	
	de	
current life cycle	en	4.2.2 Decommissioned
label	en	DORIS
	de	DORIS
tagline	en	
	de	
description	en	
	de	

09. HERA

Type	en	Circular Particle Collider
	de	Ring-Collider
current life cycle	en	4.2.2 Decommissioned
label	en	HERA
	de	HERA
tagline	en	
	de	
description	en	
	de	

10. PIA

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	PIA
	de	PIA
tagline	en	Positron intensity Accumulator
	de	Positronen-Intensitäts-Akkumulator / PETRA-Vorbeschleuniger
description	en	
	de	

11. FLASHForward

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	FLASHForward
	de	FLASHForward
tagline	en	Future-Oriented Wakefield Accelerator Research and Developement at FLASH
	de	Zukunftsweisende Plasmabeschleuniger-Forschung und -Entwicklung bei FLASH
description	en	Testinfrastruktur für teilchenstrahlgetriebene Plasma-Wakefield-Beschleunigungsexperimente, die darauf abzielen, in einigen Zentimetern ionisiertem Wasserstoff Elektronenstrahlen mit Energien von mehr als 1,5 GeV und einer ausreichenden Qualität für den Betrieb eines Freie-Elektronen-Lasers zu erzeugen
	de	Testinfrastruktur für teilchenstrahlgetriebene Plasma-Wakefield-Beschleunigungsexperimente, die darauf abzielen, in einigen Zentimetern ionisiertem Wasserstoff Elektronenstrahlen mit Energien von mehr als 1,5 GeV und einer ausreichenden Qualität für den Betrieb eines Freie-Elektronen-Lasers zu erzeugen

12. KALDERA

Type	en	Laser
	de	Laser
current life cycle	en	2.2 Construction
label	en	KALDERA
	de	KALDERA
tagline	en	High-power laser for plasma acceleration
	de	Hochleistungs-Laser für Plasma-Beschleunigung
description	en	Laser for operating a laser plasma accelerator with ultra-fast laser pulses with peak powers of 100 terawatts and repetition rates of up to 1 kHz. Project start: 2020.
	de	Laser zum Betrieb eines Laser-Plasma-Beschleunigers mit ultraschnellen Laserpulsen mit Spitzenleistungen von 100 Terawatt und Wiederholraten bis zu 1 kHz. Projektstart: 2020.

13. LUX

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	LUX
	de	LUX
tagline	en	Laser-driven plasma accelerator
	de	Laser-getriebener Plasma-Beschleuniger
description	en	Test facility for the development of laser plasma-assisted injectors for synchrotron light sources
	de	Testanlage für die Entwicklung von laserplasmagestützten Injektoren für Synchrotronlichtquellen

14. ANGUS

Type	en	Laser
	de	Laser
current life cycle	en	3.1 Operation
label	en	ANGUS
	de	ANGUS
tagline	en	Titanium-sapphire laser ANGUS
	de	Titan-Saphir-Laser ANGUS
description	en	Titanium-sapphire laser system delivering pulses of up to 5 joules of energy with a pulse duration of about 30 femtoseconds
	de	Titan-Saphir-Lasersystem, das Lichtpulse mit einer Energie von bis zu 5 Joule und einer Dauer von etwa 30 Femtosekunden liefert

15. PITZ

Type	en	Beamline
	de	Beamline
current life cycle	en	3.1 Operation
label	en	PITZ
	de	PITZ
tagline	en	Photo Injector Test Facility at Zeuthen
	de	Photo Injector Test Facility at Zeuthen
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

EUROPEAN SYNCHROTRON RADIATION FACILITY

GENERAL INFORMATION

Short label for the unit	en	ESRF
	de	ESRF
Official name of the organization	en	European Synchrotron Radiation Facility
	de	European Synchrotron Radiation Facility
One-line summary	en	Innovates with synchrotron science; operates ESRF-EBS fourth-generation storage ring.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
European Synchrotron Radiation Facility		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. ESRF

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	4.1 Upgrade
label	en	ESRF
	de	ESRF
tagline	en	European Synchrotron Radiation Facility
	de	European Synchrotron Radiation Facility
description	en	European Synchrotron Radiation Facility
	de	European Synchrotron Radiation Facility

02. ESRF

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	4.1 Upgrade
label	en	ESRF
	de	ESRF
tagline	en	European Synchrotron Radiation Facility
	de	European Synchrotron Radiation Facility
description	en	European Synchrotron Radiation Facility
	de	European Synchrotron Radiation Facility

03. ESRF-EBS

Type	en	Synchrotron Radiation Source (4th Generation)
	de	Synchrotronlichtquelle (4. Generation)
current life cycle	en	3.1 Operation
label	en	ESRF-EBS
	de	ESRF-EBS
tagline	en	European Synchrotron Radiation Facility - Extremely Brilliant source
	de	Extrem brillante Quelle
description	en	European Synchrotron Radiation Facility - Extremely Brilliant source
	de	European Synchrotron Radiation Facility - Extremely Brilliant source

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

EUROPEAN SPALLATION SOURCE

GENERAL INFORMATION

Short label for the unit	en	ESS
	de	ESS
Official name of the organization	en	European Spallation Source
	de	Europäische Spallationsquelle
One-line summary	en	Builds new accelerator technologies; hosts the ESS neutron spallation source
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
European Spallation Source		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. ESS

Type	en	Spallation Neutron Source
	de	Spallationsneutronenquelle
current life cycle	en	2.2 Construction
label	en	ESS
	de	ESS
tagline	en	European Spallation Source
	de	Europäische Spallationsquelle
description	en	European Spallation Source
	de	European Spallation Source

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

EUROPEAN X-RAY FREE-ELECTRON LASER FACILITY

GENERAL INFORMATION

Short label for the unit	en	eu.XFEL
	de	eu.XFEL
Official name of the organization	en	European X-Ray Free-Electron Laser Facility
	de	European XFEL
One-line summary	en	Develops X-ray FEL technology; operates the European XFEL
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTERSTUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
European X-Ray Free-Electron Laser Facility		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. European XFEL		
Type	en	
	de	
current life cycle	en	3. Operation And Maintenance
label	en	European XFEL
	de	European XFEL
tagline	en	European X-Ray Free-Electron Laser Facility
	de	Europäische Freie-Elektronen-Röntgenlaseranlage
description	en	
	de	

02. European XFEL		
Type	en	
	de	
current life cycle	en	3. Operation And Maintenance
label	en	European XFEL
	de	European XFEL
tagline	en	European X-Ray Free-Electron Laser Facility
	de	Europäische Freie-Elektronen-Röntgenlaseranlage
description	en	
	de	

03. European XFEL 1.5

Type	en	
	de	
current life cycle	en	1. Design
label	en	European XFEL 1.5
	de	European XFEL 1.5
tagline	en	European XFEL 1.5
	de	European XFEL 1.5
description	en	
	de	

04. European XFEL 2.0

Type	en	
	de	
current life cycle	en	1. Design
label	en	European XFEL 2.0
	de	European XFEL 2.0
tagline	en	European XFEL 2.0
	de	European XFEL 2.0
description	en	2nd fan cw operation
	de	2nd fan cw operation

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

FACILITY FOR ANTIPROTON AND ION RESEARCH

GENERAL INFORMATION

Short label for the unit	en	FAIR
	de	FAIR
Official name of the organization	en	Facility for Antiproton and Ion Research
	de	Internationale Beschleuniger-Anlage zur Forschung mit Antiprotonen und Ionen
One-line summary	en	Builds advanced ion/antiproton accelerators; hosts the future FAIR complex at GSI.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Facility for Antiproton and Ion Research		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. FAIR		
Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	FAIR
	de	FAIR
tagline	en	Facility for Antiproton and Ion Research
	de	Forschungsanlage für Antiprotonen und Ionen
description	en	
	de	:fair :fair/0001-unilac :fair/0002-p-linac :fair/1001-sis18 :fair/1002-sis100 :fair/2001-p-bar :fair/2002-super-frs :fair/3001-hesr :fair/3010-cr :fair/3011-resr :fair/3012-nesr

02. CR

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	CR
	de	CR
tagline	en	Collector Ring
	de	Speicherring
description	en	
	de	

03. HESR

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	HESR
	de	HESR
tagline	en	High-Energy Storage Ring
	de	Hochenergiespeicherring
description	en	A storage ring for antiprotons in the energy range between 0.83 GeV and 14 GeV
	de	Speicherring Antiprotonen im Energiebereich zwischen 0.83 GeV und 14 GeV

04. ESR

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	ESR
	de	NESR
tagline	en	New Experimental Storage Ring
	de	Neuer Experimentierspeicherring
description	en	
	de	

05. p-bar

Type	en	Beam target and particle separator
	de	Strahlen-Target und Teilchentrenner
current life cycle	en	2.2 Construction
label	en	p-bar
	de	p-bar
tagline	en	Target and separator for antiprotons
	de	Target und Separator für Antiprotonen
description	en	
	de	

06. p-Linac

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	p-Linac
	de	p-Linac
tagline	en	Proton linear accelerator
	de	Protonen-Linac
description	en	P-LINAC provides the protons for the international accelerator facility FAIR in Darmstadt. This accelerator consists of a proton source, an RFQ and a room temperature CH drift tube linac for accelerations of up to 70 MeV.
	de	P-LINAC liefert die Protonen für die internationale Beschleunigeranlage FAIR in Darmstadt. Dieser Beschleuniger besteht aus einer Protonenquelle, einem RFQ und einem Raumtemperatur-CH-Driftröhren-Linac für Beschleunigungen von bis zu 70 MeV.

07. RESR

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	RESR
	de	RESR
tagline	en	Recycled-Experimental-Storage-Ring
	de	Recycelter Experimentierspeicherring
description	en	
	de	

08. SIS100

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	SIS100
	de	SIS100
tagline	en	Heavy-ion synchrotron
	de	Schwerionensynchrotron
description	en	
	de	

09. SIS18

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	SIS18
	de	SIS18
tagline	en	Heavy-ion synchrotron
	de	Schwerionensynchrotron
description	en	
	de	

10. Super-FRS

Type	en	Beam target and particle separator
	de	Strahlen-Target und Teilchentrenner
current life cycle	en	2.2 Construction
label	en	Super-FRS
	de	Super-FRS
tagline	en	Super-conducting Fragment Separator
	de	Supraleitender Fragment-Trenner
description	en	The Superconducting FRagment Separator (Super-FRS) will be the most powerful in-flight separator in the world. Rare isotopes of all elements up to uranium will be produced at relativistic energies and spatially separated within a few hundred nanoseconds, allowing the study of very short-lived nuclei.
	de	Der Superconducting FRagment Separator (Super-FRS) wird der leistungsstärkste In-Flight-Separator der Welt sein. Seltene Isotope aller Elemente bis hin zum Uran werden bei relativistischen Energien erzeugt und innerhalb weniger hundert Nanosekunden räumlich getrennt, was die Untersuchung sehr kurzlebiger Kerne ermöglicht.

11. UNILAC

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	UNILAC
	de	UNILAC
tagline	en	Universal Linear Accelerator
	de	Universeller Linearbeschleuniger
description	en	UNILAC (Universal Linear Accelerator) is the name of a heavy ion accelerator at the GSI Helmholtz Centre for Heavy Ion Research in Darmstadt. The 'Universal' in the name indicates that the facility can accelerate ions of all mass numbers from hydrogen to uranium. With the help of UNILAC, the elements Bohrium (Bh, 107), Hassium (Hs, 108), Meitnerium (Mt, 109), Darmstadtium (Ds, 110), Roentgenium (Rg, 111) and Copernicium (Cn, 112), among others, were synthesised for the first time.
	de	UNILAC (Universal Linear Accelerator) ist der Name eines Schwerionenbeschleunigers am GSI Helmholtzzentrum für Schwerionenforschung in Darmstadt. Das „Universal“ im Namen weist darauf hin, dass die Anlage Ionen aller Massenzahlen von Wasserstoff bis Uran beschleunigen kann. Mit Hilfe des UNILAC wurden unter anderem erstmals die Elemente Bohrium (Bh, 107), Hassium (Hs, 108), Meitnerium (Mt, 109), Darmstadtium (Ds, 110), Roentgenium (Rg, 111) und Copernicium (Cn, 112) synthetisiert.

TEST FACILITIES

01. CRYRING@ESR

Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	CRYRING@ESR
	de	CRYRING@ESR
tagline	en	Heavy ion storage ring at FAIR
	de	Schwerionenspeicherring bei FAIR
description	en	The CRYRING@ESR heavy ion storage ring is part of the FAIR facility complex. Here, ions from the ESR can be slowed down to low energies (up to a few 100 keV/u), cooled and stored. The ring was operated at Stockholm University from 1991 to 2010. As part of Sweden's contribution to the FAIR project, the facility was modernized and set up at GSI between 2013 and 2016.
	de	Der Schwerionenspeicherring CRYRING@ESR ist Teil des FAIR-Anlagen-Komplexers. Dort können Ionen aus dem ESR auf niedrige Energien (bis zu einigen 100 keV/u) abgebremst, gekühlt und gespeichert werden. Der Ring wurde von 1991 bis 2010 an der Universität Stockholm betrieben. Im Rahmen des Beitrags Schwedens zum FAIR-Projekt wurde die Anlage zwischen 2013 und 2016 bei GSI modernisiert und aufgebaut.

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

FRAUNHOFER INSTITUTE FOR ELECTRON BEAM AND PLASMA TECHNOLOGY FEP

GENERAL INFORMATION

Short label for the unit	en	FEP
	de	FEP
Official name of the organization	en	Fraunhofer Institute for Electron Beam and Plasma Technology FEP
	de	Fraunhofer-Institut für Elektronenstrahl- und Plasmatechnik FEP
One-line summary	en	Innovates electron beam/plasma tech; operates Dresden-based electron beam and plasma labs.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Fraunhofer Institute for Electron Beam and Plasma Technology FEP		-	-	-	-	-	-	-	-	-	-	-	-	0
Electron beam		-	-	-	-	-	-	-	-	-	-	-	-	0
Customized EB Systems and Technologies		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

GSI HELMHOLTZ CENTRE FOR HEAVY ION RESEARCH

GENERAL INFORMATION

Short label for the unit	en	GSI
	de	GSI
Official name of the organization	en	GSI Helmholtz Centre for Heavy Ion Research
	de	GSI Helmholtzzentrum für Schwerionenforschung
One-line summary	en	Develops heavy-ion accelerator research; operates UNILAC, SIS18, ESR, and hosts the FAIR complex.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
GSI Helmholtz Centre for Heavy Ion Research		-	-	-	-	-	-	-	-	-	-	-	-	0
Accelerator Operations		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Control systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Applications Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Front-End Control Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Group Electronics		-	-	-	-	-	-	-	-	-	-	-	-	0
System Services Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Timing Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Networks and IT-Infrastructure Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Industrial Controls Group		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Cryogenics		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Beam Diagnostics		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Vacuum systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Linac		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Decelerators		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Accelerator physics		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: ECR		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Operation Department		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: High current group		-	-	-	-	-	-	-	-	-	-	-	-	0
Department UNILAC RF		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Electric power systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Ion source department		-	-	-	-	-	-	-	-	-	-	-	-	0
Department NC Magnets & Alignment		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Operation		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Application Support		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Technical Support		-	-	-	-	-	-	-	-	-	-	-	-	0

NAME	MEMBERS	SSOR	3HER	DOC	DEUT	DEUT	DEUT	ICIST	VEER	THER	MALE	MALE	MARY	SUM
Department Post Stripper Upgrade		-	-	-	-	-	-	-	-	-	-	-	-	0
Department p-linac/p-bar-target		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Ring HV		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Ring RF Systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Department SC Magnets & Testing		-	-	-	-	-	-	-	-	-	-	-	-	0
Department Storage Rings		-	-	-	-	-	-	-	-	-	-	-	-	0
Department SIS100/SIS18		-	-	-	-	-	-	-	-	-	-	-	-	0
Department System Design SIS100/SIS18		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Vacuum systems		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

01. LIGHT@GSI

Type	en	Seeded Free-Electron Laser
	de	Geimpfter Freie-Elektronen-Laser
current life cycle	en	3.1 Operation
label	en	LIGHT@GSI
	de	LIGHT@GSI
tagline	en	Laser Ion Generation, Handling and Transport
	de	Laser Ion Generation, Handling and Transport
description	en	
	de	

02. PHELIX

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	PHELIX
	de	PHELIX
tagline	en	PW-class laser PHELIX with dedicated plasma accelerator
	de	PW-class laser PHELIX with dedicated plasma accelerator
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HELMHOLTZ INSTITUTE JENA

GENERAL INFORMATION

Short label for the unit	en	HIJ
	de	HIJ
Official name of the organization	en	Helmholtz Institute Jena
	de	Helmholtz-Institut Jena
One-line summary	en	Focuses on laser plasma accelerators; collaborates with European XFEL, FAIR, and FLASH-based facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Helmholtz Institute Jena		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

01. POLARIS

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	2. Realisation
label	en	POLARIS
	de	POLARIS
tagline	en	Petawatt Optical Laser Amplifier for Radiation Intensive ExperimentS
	de	Petawatt-Laserverstärker für strahlungsintensive Experimente
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HELMHOLTZ INSTITUTE MAINZ

GENERAL INFORMATION

Short label for the unit	en	HIM
	de	HIM
Official name of the organization	en	Helmholtz Institute Mainz
	de	Helmholtz-Institut Mainz
One-line summary	en	Advances superconducting accelerator R&D; operates MAMI, MESA (under construction).
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Helmholtz Institute Mainz		-	-	-	-	-	-	-	-	-	-	-	-	0
Section: Accelerator Design and Integrated Detectors		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HUMBOLDT UNIVERSITY OF BERLIN

GENERAL INFORMATION

Short label for the unit	en	HU Berlin
	de	HU Berlin
Official name of the organization	en	Humboldt University of Berlin
	de	Humboldt-Universität zu Berlin
One-line summary	en	Researches accelerator science; based in Adlershof campus with access to joint Berlin scientific facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Humboldt University of Berlin		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

In academic year 2024/25, HU Berlin offered 4 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 7.

01. Das 1x1 der Beschleunigerphysik

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://agnes.hu-berlin.de/lupo/rds?state=verpublish&status=init&vmfile=no&publishid=2183...

02. Das 1x1 der Beschleunigerphysik (Übungen)

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-2
Link	https://agnes.hu-berlin.de/lupo/rds?state=verpublish&status=init&vmfile=no&publishid=2183...

03. Physik und Technik moderner Teilchenbeschleuniger (Übungen)

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://agnes.hu-berlin.de/lupo/rds?state=verpublish&status=init&vmfile=no&publishid=21213...

04. Physik und Technik moderner Teilchenbeschleuniger

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://agnes.hu-berlin.de/lupo/rds?state=verpublish&status=init&vmfile=no&publishid=21213...

DOCTORAL THESES

From 2010 onwards, a total of 18 doctoral theses have been published in the field of accelerator research at HU Berlin. Of these, 18 led to science degrees and none to engineering degrees. 3 were written by women and 15 by men.

01. Marten Koopmans (2022)

Two-Dimensional Bunch-Resolved Optical Diagnostics at BESSY II

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.18452/24232
fulltext link	https://edoc.hu-berlin.de/bitstreams/a9255389-708f-4b15-abad-329f0503797b/download
about facility	BESSY II
in affiliation with	HZB

02. Felix Armbrorst (2021)

Transverse Resonance Island Buckets at BESSY II. A new bunch separation scheme

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.18452/23851
fulltext link	https://edoc.hu-berlin.de/bitstreams/93ffa648-ee3a-416b-92e1-fc27494d2e7b/download
about facility	BESSY II
in affiliation with	HZB

03. Sophie Seidel (2021)

Design, Aufbau und Inbetriebnahme eines Bestrahlungsplatzes mit aktiver dreidimensionaler Aufbereitung des Protonenstrahls

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2021
Language	de
doi	10.18452/24138
fulltext link	https://edoc.hu-berlin.de/bitstreams/840f8283-8ec5-42fd-8ff71-3c0aaca7bac6/download
about facility	
in affiliation with	

04. Malte Titze (2020)

Space charge modeling at the integer resonance for the CERN PS and SPS

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.18452/21423
fulltext link	https://edoc.hu-berlin.de/bitstreams/6465fd18-7e19-4393-9ff4-4104665b5ad6/download
about facility	
in affiliation with	

05. Eva Panofski (2019)

Beam dynamics and limits for high brightness, high average current superconducting radiofrequency (SRF) photoinjectors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2019
Language	en
doi	10.18452/19962
fulltext link	https://edoc.hu-berlin.de/bitstreams/14982612-b7dd-43a3-9044-04c2f7b55f85/download
about facility	
in affiliation with	

06. Matthias Bromberger (2019)

Generation and characterization of a pulsed, intense neutron- and gamma-beam to be used in air cargo interrogation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
doi	10.18452/19816
fulltext link	https://edoc.hu-berlin.de/bitstreams/64548620-6e61-4b55-adbf-1d1ce41a9971/download
about facility	
in affiliation with	

07. Martin Anton Helmut Schmeißer (2019)

Photocathodes for high brightness, high average current photoelectron injectors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
doi	10.18452/20481
fulltext link	https://edoc.hu-berlin.de/bitstreams/c35447f5-983e-41c7-ba61-e975f58dedf8/download
about facility	
in affiliation with	

08. Roman **Martin** (2018)

Interaction Region Design for a 100 TeV Proton-Proton Collider

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.18452/19416
fulltext link	https://edoc.hu-berlin.de/bitstreams/12dfa1d3-dce3-44c9-b399-370643fc5084/download
about facility	FCC-eeFCC-hh
in affiliation with	CERN

09. Jens **Völker** (2018)

Development of a compact test facility for SRF photoelectron injectors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.18452/19322
fulltext link	https://edoc.hu-berlin.de/bitstreams/4cc8a159-00f3-4ab2-9dfa-717183db092b/download
about facility	
in affiliation with	

10. Christian **Koschitzki** (2017)

Injection mechanisms in laser wakefield acceleration

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.18452/17760
fulltext link	https://edoc.hu-berlin.de/bitstreams/63470fda-6368-4c61-8980-e3c163ac2159/download
about facility	
in affiliation with	

11. Stephanie Rädel (2017)

On space charge driven microbunching instability in bERLinPro

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2017
Language	en
doi	10.18452/17714
fulltext link	https://edoc.hu-berlin.de/bitstreams/98024b47-0c50-425c-ae21-2a7505315e67/download
about facility	
in affiliation with	

12. Martin Ruprecht (2016)

Calculation of coupled bunch effects in the synchrotron light source BESSY VSR

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
doi	10.18452/17446
fulltext link	https://edoc.hu-berlin.de/bitstreams/47918049-65d7-4ec0-a2c7-d4dab4886504/download
about facility	BESSY II
in affiliation with	HZB

13. Tobias Tydecks (2016)

A robinson wiggler for the metrology light source

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
doi	10.18452/17639
fulltext link	https://edoc.hu-berlin.de/bitstreams/f6a33b69-117c-4793-b0c5-5baddf773f80/download
about facility	
in affiliation with	

14. Christoph Kunert (2015)

Entwicklung eines Multi-Leaf Faraday Cups zur Strahldiagnose in der Augentumorthherapie

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
doi	10.18452/17153
fulltext link	https://edoc.hu-berlin.de/bitstreams/bc24f2d7-4870-4de7-af4a-3fe96594915e/download
about facility	
in affiliation with	

15. Terry Atkinson (2015)

Modeling of magnetic optic for the short pulse mode operation of Energy Recovery Linac based light sources

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
doi	10.18452/17300
fulltext link	https://edoc.hu-berlin.de/bitstreams/328a7e5c-d43d-412e-9feb-90a7df0ef57f/download
about facility	
in affiliation with	

16. Markus Ries (2014)

Nonlinear momentum compaction and coherent synchrotron radiation at the Metrology Light Source

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
doi	10.18452/16979
fulltext link	https://edoc.hu-berlin.de/bitstreams/c39f5248-5ca8-4e08-98b7-52f5a456ded0/download
about facility	BESSY II
in affiliation with	HZB

17. Yuriy **Petenev** (2014)

Analysis of injection and recovery schemes for ERL based light source

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
doi	10.18452/17014
fulltext link	https://edoc.hu-berlin.de/bitstreams/621b874e-dd44-4110-b2f9-7f0c7aeb5d72/download
about facility	
in affiliation with	

18. Lutz-Mathias **Hein** (2013)

CERN Linac4 - the space charge challenge

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	en
doi	10.18452/16802
fulltext link	https://edoc.hu-berlin.de/bitstreams/97ceae75-22ed-492e-a151-52b414c3e282/download
about facility	
in affiliation with	

HELMHOLTZ-ZENTRUM BERLIN

GENERAL INFORMATION

Short label for the unit	en	HZB
	de	HZB
Official name of the organization	en	Helmholtz-Zentrum Berlin
	de	Helmholtz-Zentrum Berlin für Materialien und Energie
One-line summary	en	Develops superconducting accelerator tech; operates BESSY II synchrotron (Berlin).
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS													RECURSIVE SUM
Helmholtz-Zentrum Berlin														
Division: Accelerators														
Department: High Brightness Beams														
Department: Protons for Therapy														
Department: Accelerator Operation, Development and Technology - Operation BESSY II / MLS														
Group: Accelerator Operation and Technology														
Group: Accelerator Operation and Technology														
Group: High Brightness Photon Beams														
Group: Storage Rings and Beam Physics														
Department: Science and Technology of Accelerating Systems														
Division: Scientific Instrumentation and Support														

NAME	MEMBERS										SSOR	HER	DOC	DEPT	DEPT	DEPT	ICIST	VEER	HER	MALE	MALE	MARY	SUM
Group: Undulators											3	6	12	-	-	-	10	11	-	8	7	-	21
																							
																							

FACILITIES

USER FACILITIES

01. BerlinProtonen

Type	en	Proton Beam Therapy Facility
	de	Anlage für Protonenbestrahlungstherapie
current life cycle	en	3.1 Operation
label	en	BerlinProtonen
	de	BerlinProtonen
tagline	en	Proton therapy for eye tumors
	de	Protonentherapie von Augentumoren
description	en	BerlinProtonen is a proton therapy facility specialising in the treatment of eye tumours, operated jointly by Helmholtz-Zentrum Berlin (HZB) and Charité. Proton beams with an energy of 68 MeV are used to precisely target and destroy eye tumours while sparing surrounding healthy tissue. The treatment has a success rate of over 97% in destroying eye tumours. In most cases, the eye is not only saved, but a satisfactory level of vision is maintained. More than 3,700 patients have been treated at the facility since 1998.
	de	BerlinProtonen ist eine auf die Behandlung von Augentumoren spezialisierte Protonentherapieanlage, die gemeinsam vom Helmholtz-Zentrum Berlin (HZB) und der Charité betrieben wird. Protonenstrahlen mit einer Energie von 68 MeV werden eingesetzt, um Augentumore gezielt zu zerstören und dabei das umliegende gesunde Gewebe zu schonen. Die Erfolgsquote bei der Zerstörung von Augentumoren liegt bei über 97 Prozent. In den meisten Fällen wird das Auge nicht nur gerettet, sondern es bleibt auch ein zufriedenstellendes Sehvermögen erhalten. Seit 1998 wurden in der Einrichtung mehr als 3 700 Patienten behandelt.

02. BerlinProtonen / 01-Pre-Accelerator

Type	en	Van De Graaff Accelerator
	de	Van-de-Graaff-Beschleuniger
current life cycle	en	3.1 Operation
label	en	BerlinProtonen / 01-Pre-Accelerator
	de	BerlinProtonen / 01-Vorbeschleuniger
tagline	en	
	de	
description	en	As a pre-accelerator, BerlinProtonen uses a Van de Graaff generator with a maximum potential of 5.5 million volts.
	de	Als Vorbeschleuniger verwendet BerlinProtonen einen Van-de-Graaff-Generator mit einer maximalen Spannung von 5,5 Millionen Volt.

03. BerlinProtonen / 02-Sector Cyclotron

Type	en	Sector Cyclotron
	de	Sektor-Zyklotron
current life cycle	en	3.1 Operation
label	en	BerlinProtonen / 02-Sector Cyclotron
	de	BerlinProtonen / 02-Sektorzyklotron
tagline	en	
	de	
description	en	As the main accelerator, BerlinProtonen uses a sector cyclotron with four bending magnets and two high-frequency components for a stepwise energy transfer. In this way, the protons reach a final energy of 68 million electronvolts, which corresponds to about half the speed of light.
	de	Als Hauptbeschleuniger nutzt BerlinProtonen ein Sektorenzyklotron mit vier Ablenkmagneten und zwei Hochfrequenzkomponenten für eine schrittweise Energieübertragung. Auf diese Weise erreichen die Protonen eine Endenergie von 68 Millionen Elektronenvolt, was etwa der halben Lichtgeschwindigkeit entspricht.

04. BESSY II

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	3.1 Operation
label	en	BESSY II
	de	BESSY II
tagline	en	Berlin electron storage ring for synchrotron radiation II
	de	Berliner Elektronenspeicherring für Synchrotronstrahlung II
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

05. BESSY II+

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	2. Realisation
label	en	BESSY II+
	de	BESSY II+
tagline	en	Modernization and upgrade initiative for BESSY II
	de	Modernisierungs- und Aufrüstungsinitiative für BESSY II
description	en	BESSY II Plus (BESSY II+) is a comprehensive modernization and upgrade initiative for the BESSY II synchrotron light source at Helmholtz-Zentrum Berlin. The project aims to safeguard and enhance BESSY II's scientific capabilities in the years leading up to the construction and commissioning of its successor, BESSY III, which is scheduled to begin operation around 2035. The scope of modernization for the accelerator complex includes general upgrades-such as replacing the uninterruptible power supply-as well as the introduction of advanced accelerator technologies. In a European collaboration between ALBA, DESY, and HZB, a normal-conducting 1.5 GHz higher harmonic cavity (HHC) with a HOM damper has been developed. The prototype, designed and procured by ALBA and RF-tested at HZB, has been successfully tested with beam and conditioned for user operation since autumn 2022. This technology is intended for use at BESSY III, enabling the simultaneous storage of short and long bunches. Advanced digitalization, utilizing a digital twin as a central interface, will support more efficient commissioning, startup, and operation of the facility. To improve sustainability, the replacement of an energy-intensive bending electromagnet in the transfer line between the booster and storage ring is under consideration. Given the PTB's specific requirements for the bending radiation source in metrology applications, there are also discussions about replacing two main bends of one double-bend-achromat section with permanent magnets.
	de	BESSY II Plus (BESSY II+) ist eine umfassende Modernisierungs- und Aufrüstungsinitiative für die Synchrotronlichtquelle BESSY II am Helmholtz-Zentrum Berlin. Das Projekt zielt darauf ab, die wissenschaftlichen Fähigkeiten von BESSY II in den Jahren bis zum Bau und der Inbetriebnahme des Nachfolgers BESSY III, der 2035 in Betrieb gehen soll, zu sichern und zu verbessern. Der Umfang der Modernisierung des Beschleunigerkomplexes umfasst sowohl allgemeine Verbesserungen - wie den Austausch der unterbrechungsfreien Stromversorgung - als auch die Einführung fortschrittlicher Beschleunigertechnologien. Im Rahmen einer europäischen Zusammenarbeit zwischen ALBA, DESY und dem HZB wurde ein normalleitender 1,5-GHz-Resonator (HHC) mit einem HOM-Dämpfer entwickelt. Der von ALBA entworfene und beschaffte Prototyp, der am HZB HF-getestet wurde, wird seit Herbst 2022 erfolgreich mit Strahl getestet und für den Nutzerbetrieb bereitgestellt. Diese Technologie ist für den Einsatz bei BESSY III vorgesehen und ermöglicht die gleichzeitige Speicherung von kurzen und langen Paketen. Eine fortgeschrittene Digitalisierung, die einen digitalen Zwilling als zentrale Schnittstelle nutzt, wird eine effizientere Inbetriebnahme und den Betrieb der Anlage unterstützen. Zur Verbesserung der Nachhaltigkeit wird der Ersatz eines energieintensiven Biegeelektromagneten in der Übertragungsleitung zwischen Booster und Speicherring in Erwägung gezogen. Angesichts der spezifischen Anforderungen der PTB an die Biegestrahlungsquelle in der Metrologie wird auch über den Ersatz von zwei Hauptbögen eines Doppelbogens diskutiert.

06. BESSY II/VSR

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	
label	en	BESSY II/VSR
	de	BESSY II/VSR
tagline	en	Variable pulse length storage ring
	de	Variabler Puls­längenspeicherring
description	en	
	de	

07. BESSY-II/01: Electron Gun

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	BESSY-II/01: Electron Gun
	de	BESSY-II/01: Elektronenquelle
tagline	en	BESSY-II electron gun (1st stage of the BESSY-II accelerator complex)
	de	BESSY-II Elektronenquelle (1. Stufe des BESSY-II-Beschleunigerkomplexes)
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

08. BESSY-II/02b: Microtron

Type	en	Racetrack Microtron
	de	Rennbahn-Mikrotron
current life cycle	en	4.1.2 Replaced By Upgrade
label	en	BESSY-II/02b: Microtron
	de	BESSY-II/02a: Microtron
tagline	en	BESSY-II microtron (2nd stage of the BESSY-II accelerator complex)
	de	BESSY-II Microtron (2. Stufe des BESSY-II-Beschleunigerkomplexes)
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

09. BESSY-II/02b: Linac

Type	en	
	de	
current life cycle	en	3. Operation And Maintenance
label	en	BESSY-II/02b: Linac
	de	BESSY-II/02b: Linac
tagline	en	BESSY-II linac (2nd stage of the BESSY-II accelerator complex)
	de	BESSY-II-Linac (2. Stufe des BESSY-II-Beschleunigerkomplexes)
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

10. BESSY-II/03: Booster Synchrotron

Type	en	Booster Synchrotron
	de	Booster-Synchrotron
current life cycle	en	3. Operation And Maintenance
label	en	BESSY-II/03: Booster Synchrotron
	de	BESSY-II/03: Booster-Synchrotron
tagline	en	BESSY-II booster synchrotron (3rd stage of the BESSY-II accelerator complex)
	de	BESSY-II-Booster-Synchrotron (3. Stufe des BESSY-II-Beschleunigerkomplexes)
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

11. BESSY-II/04: Storage Ring

Type	en	
	de	
current life cycle	en	3. Operation And Maintenance
label	en	BESSY-II/04: Storage Ring
	de	BESSY-II/04: Speicherring
tagline	en	BESSY-II storage ring (final stage of the BESSY-II accelerator complex)
	de	BESSY-II-Speicherring (Endstufe des BESSY-II-Beschleunigerkomplexes)
description	en	Storage ring for electrons in an energy range of 0.9 to 1.9 billion electron volts and a circumference of 240 metres.
	de	Speicherring für Elektronen in einem Energiebereich von 0,9 bis 1,9 Milliarden Elektronenvolt und einem Umfang von 240 Meter.

12. BESSY II

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	3.1 Operation
label	en	BESSY II
	de	BESSY II
tagline	en	Berlin electron storage ring for synchrotron radiation II
	de	Berliner Elektronenspeicherring für Synchrotronstrahlung II
description	en	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung
	de	Synchrotron-Strahlungsquelle der dritten Generation für weiche Röntgenstrahlung

13. BESSY III

Type	en	Synchrotron Radiation Source (4th Generation)
	de	Synchrotronlichtquelle (4. Generation)
current life cycle	en	1.2.1 Technical Design Creation
label	en	BESSY III
	de	BESSY III
tagline	en	Berlin electron storage ring for synchrotron radiation III
	de	Berliner Elektronenspeicherring für Synchrotronstrahlung III
description	en	Geplante Synchrotron-Strahlungsquelle der vierten Generation für weiche Röntgenstrahlung
	de	Geplante Synchrotron-Strahlungsquelle der vierten Generation für weiche Röntgenstrahlung

14. MLS

Type	en	Synchrotron Radiation Source
	de	Synchrotronlichtquelle
current life cycle	en	3. Operation And Maintenance
label	en	MLS
	de	MLS
tagline	en	Metrology Light Source
	de	Metrology Light Source
description	en	Storage ring for electrons with energies between 105 million and 630 million electronvolts and a circumference of 48 metres. The MLS stores beam intensities ranging from a single electron to tens of billions of electrons, with the intensity of the emitted radiation covering the corresponding range. Since the properties of the emitted radiation can be calculated from first principles, the MLS serves as a „standard candle“ for the calibration of other sources or spectrometers.
	de	Speicherring für Elektronen mit Energien zwischen 105 Millionen und 630 Millionen Elektronenvolt und einem Umfang von 48 Metern. Der MLS speichert Strahlungsintensitäten von einem einzelnen Elektron bis zu mehreren zehn Milliarden Elektronen, wobei die Intensität der emittierten Strahlung den entsprechenden Bereich abdeckt. Da die Eigenschaften der emittierten Strahlung nach ersten Prinzipien berechnet werden können, dient die MLS als „Standardkerze“ für die Kalibrierung anderer Quellen oder Spektrometer.

TEST FACILITIES

01. bERLinPro

Type	en	Energy Recovery Linac
	de	Linearbeschleuniger mit Energierückgewinnung
current life cycle	en	4.2.2 Decommissioned
label	en	bERLinPro
	de	bERLinPro
tagline	en	Berlin Energy Recovery LINac PROject
	de	Berliner Energierückgewinnung LINac PROjekt
description	en	Testanlage zur Elektronenbeschleunigung mit hohen Strömen im Dauerstrichbetrieb und mit Energierückgewinnung auf der Basis von supraleitenden Hohlräumen
	de	Testanlage zur Elektronenbeschleunigung mit hohen Strömen im Dauerstrichbetrieb und mit Energierückgewinnung auf der Basis von supraleitenden Hohlräumen

02. Supralab

Type	en	Facility for cryogenic measurements
	de	Anlage für Kältemessungen
current life cycle	en	3. Operation And Maintenance
label	en	Supralab
	de	Supralab
tagline	en	Test infrastructure for accelerator devices, superconducting radio-frequency systems and cryogenic applications
	de	Testinfrastruktur für Beschleuniger, supraleitende Hochfrequenzsysteme und kryogene Anwendungen
description	en	Testinfrastruktur für supraleitende Hochfrequenzsysteme
	de	Testinfrastruktur für supraleitende Hochfrequenzsysteme

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

JÜLICH RESEARCH CENTRE

GENERAL INFORMATION

Short label for the unit	en	FZJ
	de	FZJ
Official name of the organization	en	Jülich Research Centre
	de	Forschungszentrum Jülich
One-line summary	en	Advances research with COSY synchrotron and hosts accelerator and nuclear physics labs.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Jülich Research Centre		-	-	-	-	-	-	-	-	-	-	-	-	0
Neutron research		-	-	-	-	-	-	-	-	-	-	-	-	0
Engineering and Technology		-	-	-	-	-	-	-	-	-	-	-	-	0
Accelerator physics		-	-	-	-	-	-	-	-	-	-	-	-	0
Jülich Centre for Neutron Science		-	-	-	-	-	-	-	-	-	-	-	-	0
Quantum Materials and Collective Phenomena		-	-	-	-	-	-	-	-	-	-	-	-	0
High Brilliance Neutron Source		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. COSY														
Type	en													
	de													
current life cycle	en	3. Operation And Maintenance												
label	en	COSY												
	de	COSY												
tagline	en	Cooler-Synchrotron												
	de	Kühlersynchrotron												
description	en	Kühlersynchrotron für Protonen und Deuteronen am Forschungszentrum Jülich. Die Teilchenstrahlen können bei Bedarf gekühlt werden, um die Strahlqualität zu erhöhen.												
	de	Kühlersynchrotron für Protonen und Deuteronen am Forschungszentrum Jülich. Die Teilchenstrahlen können bei Bedarf gekühlt werden, um die Strahlqualität zu erhöhen.												

02. HBNS

Type	en	Compact Accelerator Neutron Source
	de	Kompakte Beschleuniger-Neutronenquelle
current life cycle	en	1. Design
label	en	HBNS
	de	HBNS
tagline	en	High Brilliance Neutron Source
	de	Hochbrillante Neutronenquelle
description	en	High Brilliance Neutron Source
	de	High Brilliance Neutron Source

TEST FACILITIES

01. JuSPARC

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	JuSPARC
	de	JuSPARC
tagline	en	Jülich Short-Pulsed Particle and Radiation Center
	de	Jülicher Zentrum für kurzgepulste Teilchen und Strahlung
description	en	Laser-based test facility for accelerator research with short-pulsed photon and particle beams
	de	Laserbasierte Testanlage für Beschleunigerforschung mit kurzgepulsten Photonen- und Teilchenstrahlen

02. JuSPARC VEGA

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	JuSPARC VEGA
	de	JuSPARC VEGA
tagline	en	JuSPARC VEGA
	de	JuSPARC VEGA
description	en	
	de	

03. JuSPARC SIRIUS

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	JuSPARC SIRIUS
	de	JuSPARC SIRIUS
tagline	en	JuSPARC SIRIUS
	de	JuSPARC SIRIUS
description	en	
	de	

04. JuSPARC MIRA

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	JuSPARC MIRA
	de	JuSPARC MIRA
tagline	en	JuSPARC MIRA
	de	JuSPARC MIRA
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HELMHOLTZ-ZENTRUM HEREON

GENERAL INFORMATION

Short label for the unit	en	HZG
	de	HZG
Official name of the organization	en	Helmholtz-Zentrum Hereon
	de	Helmholtz-Zentrum Hereon
One-line summary	en	Applies accelerator-based science; operates neutron and photon facilities for materials/environmental research.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Helmholtz-Zentrum Hereon		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

KARLSRUHE INSTITUTE OF TECHNOLOGY

GENERAL INFORMATION

Short label for the unit	en	KIT
	de	KIT
Official name of the organization	en	Karlsruhe Institute of Technology
	de	Karlsruher Institut für Technologie
One-line summary	en	Pioneers sustainable and flexible particle accelerator systems with advanced technologies.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Karlsruhe Institute of Technology		3	17	4	7	-	-	30	1	-	2	29	-	31
Division V: Physics and Mathematics		3	17	4	7	-	-	30	1	-	2	29	-	31
Institute for Beam Physics and Technology		3	17	4	7	-	-	30	1	-	2	29	-	31
Department: Accelerator Research and Development + Operations I		1	9	1	2	-	-	13	-	-	1	12	-	13
Group: Accelerators		-	8	1	2	-	-	11	-	-	-	11	-	11
Team: Facilities	    	-	4	1	-	-	-	5	-	-	-	5	-	5
Team: Beam Dynamics & Theory	   	-	2	-	2	-	-	4	-	-	-	4	-	4
Team: Controls & Instrumentation		-	1	-	-	-	-	1	-	-	-	1	-	1
Applications THz/IR		-	1	-	-	-	-	1	-	-	1	-	-	1
Research Group Supraleitende Systemtechnologie	 	-	-	-	2	-	-	2	-	-	-	2	-	2
Department: Accelerator Research and Development + Operations II		1	8	3	3	-	-	14	1	-	-	15	-	15
Group: New Accelerator Concepts		-	1	1	-	-	-	2	-	-	-	2	-	2
Team: NAC		-	-	1	-	-	-	1	-	-	-	1	-	1
Group: Insertion Devices		1	2	-	1	-	-	3	1	-	-	4	-	4
Team: Grau		-	1	-	-	-	-	1	-	-	-	1	-	1
Team: Saez de Jauregui	 	-	1	-	1	-	-	2	-	-	-	2	-	2
Group: Diagnosis R&D		-	4	2	2	-	-	8	-	-	-	8	-	8
Team: AL4Accelerators		-	-	-	-	-	-	-	-	-	-	-	-	0
Team: Funker	 	-	1	-	1	-	-	2	-	-	-	2	-	2
Team: Gethmann		-	-	1	-	-	-	1	-	-	-	1	-	1
Team: Niehues	 	-	1	1	-	-	-	2	-	-	-	2	-	2
Team: Nasse		-	1	-	-	-	-	1	-	-	-	1	-	1
Team: Steinmann	 	-	1	-	1	-	-	2	-	-	-	2	-	2

FACILITIES

USER FACILITIES

TEST FACILITIES

01. ANKA

Type	en	Synchrotron Radiation Source (3rd Generation)
	de	Synchrotronlichtquelle (3. Generation)
current life cycle	en	4.1.2 Replaced By Upgrade
label	en	ANKA
	de	ANKA
tagline	en	Angströmquelle Karlsruhe
	de	Angströmquelle Karlsruhe
description	en	
	de	

02. KIT ATP

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	KIT ATP
	de	KIT ATP
tagline	en	KIT Accelerator Technology Platform
	de	KIT Accelerator Technology Platform
description	en	
	de	

03. cSTART

Type	en	
	de	
current life cycle	en	2.1 Preparation
label	en	cSTART
	de	cSTART
tagline	en	Compact Storage Ring for Accelerator Research and Technology
	de	Kompakter Speicherring für Beschleunigerforschung und -technologie
description	en	Machbarkeitsstudie zur Speicherung ultrakurzer Elektronenpakete mit großer Impulsspanne aus einem Laser-Teilchenbeschleuniger.
	de	Machbarkeitsstudie zur Speicherung ultrakurzer Elektronenpakete mit großer Impulsspanne aus einem Laser-Teilchenbeschleuniger.

04. FLUTE

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	FLUTE
	de	FLUTE
tagline	en	Linear Accelerator Test Facility
	de	Linear-Beschleuniger-Testanlage
description	en	Elektronenbeschleuniger zur Erzeugung ultrakurzer Elektronenpakete, der aus einem femtosekundenlaser-getriebenen Photoinjektor, einem Linearbeschleuniger und einem Elektronenpakete-Kompressor besteht.
	de	Elektronenbeschleuniger zur Erzeugung ultrakurzer Elektronenpakete, der aus einem femtosekundenlaser-getriebenen Photoinjektor, einem Linearbeschleuniger und einem Elektronenpakete-Kompressor besteht. FLUTE steht für Ferninfrarot Linac- und Test-Experiment.

05. KARA

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	KARA
	de	KARA
tagline	en	Karlsruhe Research Accelerator KARA
	de	Karlsruhe Research Accelerator KARA
description	en	The Karlsruhe Research Accelerator (KARA) is a storage ring for electron beams in the energy range of 500 to 2,500 MeV, used for accelerator research and as a source of intense electromagnetic synchrotron radiation.
	de	Der Karlsruhe Research Accelerator (KARA) ist ein Speicherring für Elektronenstrahlen im Energiebereich von 500 bis 2.500 MeV, der für die Beschleunigerforschung und als Quelle für intensive elektromagnetische Synchrotronstrahlung genutzt wird.

06. KARA/01: Microtron

Type	en	Circular Microtron
	de	Kreismikrotron
current life cycle	en	3.1 Operation
label	en	KARA/01: Microtron
	de	KARA/01: Mikrotron
tagline	en	KARA-Microtron (1st stage of the KARA accelerator complex)
	de	KARA-Mikrotron (1. Stufe des KARA-Beschleunigerkomplexes)
description	en	The first stage of pre-acceleration at KARA consists of a microtron that accelerates electrons from 0.093 MeV to 53 MeV.
	de	Das Kara-Mikrotron ist der erste der beiden Vorbeschleuniger für den KARA-Speicherring, mit dem Elektronen aus einer Elektronenquelle von 0,093 MeV auf 53 MeV beschleunigt werden.

07. KARA/02: Booster Synchrotron

Type	en	Booster Synchrotron
	de	Booster-Synchrotron
current life cycle	en	3.1 Operation
label	en	KARA/02: Booster Synchrotron
	de	KARA/02: Booster-Synchrotron
tagline	en	KARA booster synchrotron (2nd stage of the KARA accelerator complex)
	de	KARA Booster-Synchrotron (2. Stufe des KARA-Beschleunigerkomplexes)
description	en	The Karlsruhe Research Accelerator (KARA) is a storage ring for electron beams in the energy range of 500 to 2,500 MeV, used for accelerator research and as a source of intense electromagnetic synchrotron radiation.
	de	Der Karlsruhe Research Accelerator (KARA) ist ein Speicherring für Elektronenstrahlen im Energiebereich von 500 bis 2.500 MeV, der für die Beschleunigerforschung und als Quelle für intensive elektromagnetische Synchrotronstrahlung genutzt wird.

08. KARA/03: Storage Ring

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	KARA/03: Storage Ring
	de	KARA/03: Speicherring
tagline	en	Karlsruhe Research Accelerator KARA
	de	KARA-Speicherring (Endstufe des KARA-Beschleunigerkomplexes)
description	en	The Karlsruhe Research Accelerator (KARA) is a storage ring for electron beams in the energy range of 500 to 2,500 MeV, used for accelerator research and as a source of intense electromagnetic synchrotron radiation.
	de	Der Karlsruhe Research Accelerator (KARA) ist ein Speicherring für Elektronenstrahlen im Energiebereich von 500 bis 2.500 MeV, der für die Beschleunigerforschung und als Quelle für intensive elektromagnetische Synchrotronstrahlung genutzt wird.

09. MCF

Type	en	Facility for magnet measurements
	de	Anlage für Magnetvermessung
current life cycle	en	3.1 Operation
label	en	MCF
	de	MCF
tagline	en	Magnet Characterization Facilities
	de	Magnet Characterization Facilities
description	en	Das MCF umfasst Prüfstände für supraleitende, normalleitende und permanentmagnetische Systeme bei Raumtemperatur, in flüssigem Helium (LHe) und ohne Kryogen.
	de	Das MCF umfasst Prüfstände für supraleitende, normalleitende und permanentmagnetische Systeme bei Raumtemperatur, in flüssigem Helium (LHe) und ohne Kryogen.

10. CASPER I

Type	en	Facility for cryogenic measurements
	de	Anlage für Kältemessungen
current life cycle	en	3.1 Operation
label	en	CASPER I
	de	CASPER I
tagline	en	
	de	
description	en	Test facility where coils with maximum dimensions of 35 cm length and 30 cm diameter can be immersed in liquid helium for testing (quench tests and local field measurements)
	de	Testanlage, in der Spulen mit maximalen Abmessungen von 35 cm Länge und 30 cm Durchmesser zu Testzwecken in flüssiges Helium getaucht werden können (Quench-Tests und lokale Feldmessungen)

11. CASPER II

Type	en	Facility for cryogenic measurements
	de	Anlage für Kältemessungen
current life cycle	en	3.1 Operation
label	en	CASPER II
	de	CASPER II
tagline	en	
	de	
description	en	Horizontal, cryogen-free test stand for quality certification of new superconducting systems and insertion devices up to 2 metres in length.
	de	Horizontaler, kryogenfreier Prüfstand für die Qualitätszertifizierung von neuen supraleitenden Systemen und Einschubgeräten bis zu 2 Metern Länge.

12. LASMAGLab

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	LASMAGLab
	de	LASMAGLab
tagline	en	
	de	
description	en	State-of-the-art measurement bench for normal conducting and permanent accelerator magnets that integrates multiple techniques such as 3D Hall probe scans, stretched wire and rotating coil field integral measurements.
	de	Messplatz nach neuestem Stand der Technik für normalleitende und permanente Beschleunigermagnete, der mehrere Techniken wie 3D-Hallsonden-Scans, Messungen mit gestrecktem Draht und Feldintegralmessungen mit rotierender Spule integriert.

TEACHING

TEACHING OFFERS

In academic year 2024/25, KIT offered 6 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 14.

01. Accelerators and Synchrotron Radiation for Materials Research with Tutorials and a Practical Training

Teaching event type	Block Course
Weekly semester hours	5
Languages	English
Semesters	2024-1
Link	https://campus.studium.kit.edu/ev/WNs-nl92TAqXKYhnYBkHrQ/de

02. Accelerators and Detectors - Future Technologies for Research and Medicine

Teaching event type	Graduate Seminar
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://campus.studium.kit.edu/ev/JqBh6KZqRw-IW0dTJ-slgQ/de

03. Modern Accelerators and Research with Photons

Teaching event type	Graduate Seminar
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://campus.studium.kit.edu/ev/3GPhUnhPSmSA9g7f1Tytg/de

04. Practical Exercises at KARA for Accelerator Physics

Teaching event type	Lab Practical
Weekly semester hours	1
Languages	German
Semesters	2024-1
Link	https://campus.studium.kit.edu/ev/_5YRyK5CQP2XkCqJjpqKVw/de

05. Particle Accelerator Physics

Teaching event type	Lecture
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://campus.studium.kit.edu/ev/1YSKJCOATOWjyfwLTcXyVq/de

06. Beschleunigerphysik (Tutorial)

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://campus.studium.kit.edu/ev/1YSKJCOATOWjyfwLTcXyVg/de

DOCTORAL THESES

From 2010 onwards, a total of 35 doctoral theses have been published in the field of accelerator research at KIT. Of these, 18 led to science degrees and 17 to engineering degrees. 10 were written by women and 25 by men.

01. Jens Tamson (2024)

Apparatus for the measurement of physical properties of cryogenic fluid mixtures

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.5445/IR/1000171277
fulltext link	https://publikationen.bibliothek.kit.edu/1000171277/153048582
about facility	
in affiliation with	

02. Sebastian Maier (2024)

Systematic Studies of the Micro-Bunching Instability with an Additional Corrugated Structure Impedance at KARA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.5445/IR/1000171277
fulltext link	https://publikationen.bibliothek.kit.edu/1000171277/153048582
about facility	KARA
in affiliation with	

03. Samira Fatehi (2023)

Compact High-Temperature Superconducting Magnets for Laser-Plasma Accelerator Beam Capture and Transport

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2023
Language	en
doi	10.5445/IR/1000158431
fulltext link	https://publikationen.bibliothek.kit.edu/1000158431/150685764
about facility	
in affiliation with	

04. Meghana Mahaveer Patil (2023)

Development and Integration of High Throughput Detector Systems for Photon-Based Diagnostics at Karlsruhe Research Accelerator

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2023
Language	en
doi	10.5445/IR/1000163059
fulltext link	https://publikationen.bibliothek.kit.edu/1000163059/151530886
about facility	KARA
in affiliation with	

05. Alexander Marek (2022)

New Type of Sub-Thz Oscillator and Amplifier Systems Based on Helical-Type Gyro-Twts

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2022
Language	en
doi	10.5445/IR/1000150852
fulltext link	https://publikationen.bibliothek.kit.edu/1000150852/149359711
about facility	
in affiliation with	

06. Eugen Shabagin (2022)

Development of a CMRC cooled 10 kA current lead for HTS applications

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2022
Language	en
doi	10.5445/IR/1000144514
fulltext link	https://publikationen.bibliothek.kit.edu/1000144514/148619284
about facility	
in affiliation with	

07. Patrick Schreiber (2022)

Negative momentum compaction operation and its effect on the beam dynamics at the accelerator test facility KARA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.5445/IR/1000148354
fulltext link	https://publikationen.bibliothek.kit.edu/1000148354/149011844
about facility	KARA
in affiliation with	

08. Kantaphon Damminsek (2022)

Magnetic characterization studies of a superconducting transverse gradient undulator for a compact LWFA-based free-electron laser

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.5445/IR/1000153441
fulltext link	https://publikationen.bibliothek.kit.edu/1000153441/149833381
about facility	
in affiliation with	

09. Tobias Boltz (2021)

Micro-Bunching Control at Electron Storage Rings with Reinforcement Learning

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.5445/IR/1000140271
fulltext link	https://publikationen.bibliothek.kit.edu/1000140271/133990608
about facility	
in affiliation with	

10. Christina Weber (2021)

Dynamic modelling of incidents for the protection of helium cryostats against excessive pressure

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2021
Language	en
doi	10.5445/IR/1000132285
fulltext link	https://publikationen.bibliothek.kit.edu/1000132285/112992329
about facility	
in affiliation with	

11. Johannes David Gomse (2020)

Development of Heat Exchanger Technology for Cryogenic Mixed Refrigerant Cycles

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2020
Language	en
about facility	
in affiliation with	

12. Andreas Janzen (2020)

Entwicklung eines neuen Messverfahrens zur Durchflussmessung in der Kryotechnik

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2020
Language	de
isbn	978-3-8439-4680-3
about facility	
in affiliation with	

13. Thomas Kochenburger (2019)

Kryogene Gemischkältekreisläufe für Hochtemperatursupraleiter-Anwendungen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2019
Language	de
isbn	978-3-8439-3987-4
about facility	
in affiliation with	

14. Daniel Berkowitz Zamora (2019)

Development of a cryogenic refrigerator for a compact superconducting cyclotron in a hospital environment

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2019
Language	en
doi	10.5445/IR/1000099481
fulltext link	https://publikationen.bibliothek.kit.edu/1000099481/46872294
about facility	
in affiliation with	

15. Prasoon Raj (2019)

Development and Testing of Self–Powered Detectors for Nuclear Measurements in Fusion Reactors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2019
Language	en
doi	10.5445/IR/1000096884
fulltext link	https://publikationen.bibliothek.kit.edu/1000096884/38644300
about facility	
in affiliation with	

16. Benjamin Kehrer (2019)

Time-resolved studies of the micro-bunching instability at KARA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2019
Language	en
doi	10.5445/IR/1000098584
fulltext link	https://publikationen.bibliothek.kit.edu/1000098584/42245503
about facility	ELBEKARA
in affiliation with	

17. Juliane Raasch (2019)

Electrical-Field Sensitive Yba2cu3o(7-X) Detectors for Real-Time Monitoring of Picosecond Thz Pulses

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2019
Language	en
doi	10.5445/KSP/1000081476
isbn	978-3-7315-0786-4
fulltext link	https://publikationen.bibliothek.kit.edu/1000081476/25411195
about facility	
in affiliation with	

18. Dora Szalkai (2019)

Experimental investigations for development of an innovative SiC sensor for spectrometry of neutrons under harsh environmental conditions

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2019
Language	en
doi	10.5445/IR/1000096850
fulltext link	https://publikationen.bibliothek.kit.edu/1000096850/35992424
about facility	
in affiliation with	

19. Carolin Zoller (2018)

Experimental Investigation and Modelling of Incidents in Liquid Helium Cryostats

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2018
Language	en
doi	10.5445/IR/1000082999
fulltext link	https://publikationen.bibliothek.kit.edu/1000082999/13655758
about facility	
in affiliation with	

20. Lorenzo Rota (2018)

KALYPSO, a novel detector system for high-repetition rate and real-time beam diagnostics

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
doi	10.5445/IR/1000071240
fulltext link	https://publikationen.bibliothek.kit.edu/1000071240/4215132
about facility	
in affiliation with	

21. Johannes Steinmann (2018)

Diagnostics of short electron bunches with THz detectors in particle accelerators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
doi	10.5445/KSP/1000090017
fulltext link	https://publikationen.bibliothek.kit.edu/1000090017/32835641
about facility	
in affiliation with	

22. Patrick Schönfeldt (2018)

Simulation and measurement of the dynamics of ultra-short electron bunch profiles for the generation of coherent THz radiation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.5445/IR/1000084466
fulltext link	https://publikationen.bibliothek.kit.edu/1000084466/15923627
about facility	
in affiliation with	

23. Bert Green (2017)

Superradiant Terahertz Sources and Their Applications in Accelerator Diagnostics and Ultra-Fast Science

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.5445/IR/1000070730
fulltext link	https://publikationen.bibliothek.kit.edu/1000070730/4204814
about facility	
in affiliation with	

24. Bastian Härer (2017)

Lattice Design and Beam Optics Calculations for the New Large-Scale Electron-Positron Collider FCC-ee

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.5445/IR/1000071240
fulltext link	https://publikationen.bibliothek.kit.edu/1000071240/4215132
about facility	FCC-ee
in affiliation with	

25. Cheng Chang (2016)

Precise determination of the electron beam energy with Compton backscattered laser photons at ANKA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
doi	10.5445/IR/1000051914
fulltext link	https://publikationen.bibliothek.kit.edu/1000051914/3785770
about facility	ANKAKARA
in affiliation with	

26. Christina Widmann (2016)

Simulation and First Experimental Tests of an Electron Beam Transport System for a Laser Wakefield Accelerator

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2016
Language	en
doi	10.5445/IR/1000055008
fulltext link	https://publikationen.bibliothek.kit.edu/1000055008/3849739
about facility	
in affiliation with	

27. Robert Voutta (2016)

Beam heat load investigations with a cold vacuum chamber for diagnostics in a synchrotron light source

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
doi	10.5445/IR/1000054479
fulltext link	https://publikationen.bibliothek.kit.edu/1000054479/3842103
about facility	
in affiliation with	

28. Veronica Afonso-Rodrigues (2015)

Electromagnetic Design, Implementation and Test of a Superconducting Undulator with a Transverse Gradient Field Amplitude

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2015
Language	en
doi	10.5445/IR/1000050911
fulltext link	https://publikationen.bibliothek.kit.edu/1000050911/3765207
about facility	
in affiliation with	

29. Miriam Fitterer (2013)

Design Study of the Large Hadron Electron Collider and a Rapid Cycling Synchrotron as Alternative to the PS Booster Upgrade at {CERN}

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2013
Language	en
doi	10.5445/IR/1000035142
fulltext link	https://publikationen.bibliothek.kit.edu/1000035142/2624738
about facility	LHC PS
in affiliation with	

30. Stefan Gerstl (2013)

A COLD vacuum chamber for beam heat load DIAGnostics (COLDDIAG)

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	en
doi	10.5445/IR/1000038354
fulltext link	https://publikationen.bibliothek.kit.edu/1000038354/2969167
about facility	
in affiliation with	

31. Steffen Hillenbrand (2013)

Study of Plasma-Based Acceleration for High Energy Physics and Other Applications

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	en
doi	10.5445/IR/1000042020
fulltext link	https://publikationen.bibliothek.kit.edu/1000042020/3152531
about facility	
in affiliation with	

32. Nicole Hiller (2013)

Electro-optical bunch length measurements at the ANKA storage ring

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2013
Language	en
doi	10.5445/IR/1000041159
fulltext link	https://publikationen.bibliothek.kit.edu/1000041159/3096634
about facility	ANKA
in affiliation with	

33. Vitali Judin (2013)

Untersuchung von Bunch-Bunch-Wechselwirkungen und des Einflusses der geometrischen Impedanz bei der Erzeugung kohärenter THz-Strahlung

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2013
Language	de
doi	10.5445/IR/1000039528
fulltext link	https://publikationen.bibliothek.kit.edu/1000039528/3013281
about facility	
in affiliation with	

34. Marit Klein (2012)

Optics calculations and simulations of longitudinal beam dynamics for the low-alpha mode at ANKA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2012
Language	en
about facility	ANKA
in affiliation with	

35. André Hofmann (2011)

Erste Untersuchungen mit einzelnen Elektronenpulsen am ANKA-Speicherring

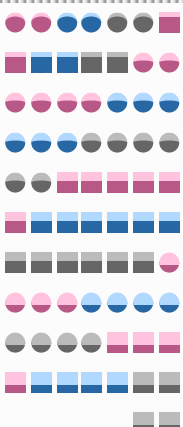
Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	de
doi	10.5445/IR/1000025363
fulltext link	https://publikationen.bibliothek.kit.edu/1000025363/1949834
about facility	ANKA
in affiliation with	

HELMHOLTZ-ZENTRUM DRESDEN-ROSSENDORF

GENERAL INFORMATION

Short label for the unit	en	HZDR
	de	HZDR
Official name of the organization	en	Helmholtz-Zentrum Dresden-Rossendorf
	de	Helmholtz-Zentrum Dresden-Rossendorf
One-line summary	en	Explores ion/electron accelerator research; operates ELBE Center for High-Power Radiation Sources.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Helmholtz-Zentrum Dresden-Rossendorf		-	12	36	24	-	-	36	36	-	24	24	-	72
Institute of Radiooncology		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: High-precision proton therapy		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Medical radiation physics		-	-	-	-	-	-	-	-	-	-	-	-	0
Institute for Radiopharmaceutical Cancer Research		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Targetry, target chemistry and radiopharmacy		-	-	-	-	-	-	-	-	-	-	-	-	0
Institute for Radiation Physics		-	12	36	24	-	-	36	36	-	24	24	-	72
Department: High Energy Density		-	12	36	24	-	-	36	36	-	24	24	-	72
Department: Radiation Source ELBE		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Helmholtz International Beamline for Extreme Fields		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: HIBEF 2.0		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Construction and commissioning of HIBEF		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: HIBEF Laser Systems		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Strong-field QED		-	-	-	-	-	-	-	-	-	-	-	-	0
Department of Nuclear Physics		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Detector development for basic research		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Nuclear physics data for science and technology		-	-	-	-	-	-	-	-	-	-	-	-	0

NAME	MEMBERS	SSOR	HER	DOC	DEUT	DEUT	DEUT	ICIST	VEER	HER	MALE	MALE	MARY	SUM
Department: Nuclear Astrophysics		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Positron annihilation spectroscopy		-	-	-	-	-	-	-	-	-	-	-	-	0
Department of Laser Particle Acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Application-orientated laser particle acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Computer-aided radiation physics		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: High power laser DRACO		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Laser-electron acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Laser ion acceleration		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Diodengepumpter PW Laser PENELOPE		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. ELBE

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	ELBE
	de	ELBE
tagline	en	User Facility ELBE
	de	Strahlungsquelle ELBE
description	en	The ELBE radiation source (electron linear accelerator for high brilliance and low emittance beams) consists of a superconducting electron accelerator and several auxiliary facilities that generate both electromagnetic radiation (THz, infrared and high-energy gamma radiation) and particle radiation (neutrons, positrons and electrons). This secondary radiation is available to both external users and members of the Helmholtz-Zentrum Dresden-Rossendorf for research purposes.
	de	Die Strahlungsquelle ELBE (Elektronen-Linearbeschleuniger für Strahlen hoher Brillanz und niedriger Emittanz) besteht aus einem supraleitenden Elektronenbeschleuniger und mehreren Nebenanlagen, die sowohl elektromagnetische Strahlung (THz-, Infrarot- und hochenergetische Gammastrahlung) als auch Teilchenstrahlung (Neutronen, Positronen und Elektronen) erzeugen. Diese Sekundärstrahlung steht sowohl externen Nutzern als auch den Mitgliedern des Helmholtz-Zentrums Dresden-Rossendorf für Forschungszwecke zur Verfügung.

TEST FACILITIES

01. DALI

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	DALI
	de	DALI
tagline	en	Dresden Advanced Light Infrastructure
	de	Fortschrittliche Lichtinfrastruktur in Dresden
description	en	Dresden Advanced Light Infrastructure
	de	Dresden Advanced Light Infrastructure

02. DRACO

Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	4.2.2 Decommissioned
label	en	DRACO
	de	DRACO
tagline	en	DResden laser ACceleration source
	de	DResden laser ACceleration source
description	en	Ultrakurzpuls-Hochleistungslaser für Untersuchungen zur Laserteilchenbeschleunigung
	de	Ultrakurzpuls-Hochleistungslaser für Untersuchungen zur Laserteilchenbeschleunigung

03. IBC

Type	en	Ion Beam Facility
	de	Ionenstrahl-Anlage
current life cycle	en	3.1 Operation
label	en	IBC
	de	IBC
tagline	en	Ion Beam Center
	de	Ionenstrahlzentrum
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be

displayed here.

MAX PLANCK INSTITUTE FOR NUCLEAR PHYSICS

GENERAL INFORMATION

Short label for the unit	en	MPIK
	de	MPIK
Official name of the organization	en	Max Planck Institute for Nuclear Physics
	de	Max-Planck-Institut für Kernphysik
One-line summary	en	Conducts accelerator-based fundamental physics; operates underground beam and detector facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Max Planck Institute for Nuclear Physics		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

GERMAN NATIONAL METROLOGY INSTITUTE

GENERAL INFORMATION

Short label for the unit	en	PTB
	de	PTB
Official name of the organization	en	German National Metrology Institute
	de	Physikalisch-Technische Bundesanstalt
One-line summary	en	Develops accelerator-based metrology; operates electron and ion accelerators for scientific standards.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
German National Metrology Institute		-	-	-	-	-	-	-	-	-	-	-	-	0
Division 7: Temperature and Synchrotron Radiation		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: Radiometry with Synchrotron Radiation		-	-	-	-	-	-	-	-	-	-	-	-	0
Department: X-ray Metrology with Synchrotron Radiation		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

TU DORTMUND UNIVERSITY

GENERAL INFORMATION

Short label for the unit	en	TU Dortmund
	de	TU Dortmund
Official name of the organization	en	TU Dortmund University
	de	Technische Universität Dortmund
One-line summary	en	Researches synchrotron and accelerator physics; operates DELTA synchrotron source.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
TU Dortmund University		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

01. DELTA		
Type	en	
	de	
current life cycle	en	3. Operation And Maintenance
label	en	DELTA
	de	DELTA
tagline	en	Dortmund ELection storage ring facility
	de	Dortmunder Elektronenspeicherring-Anlage
description	en	
	de	

TEACHING

TEACHING OFFERS

In academic year 2024/25, TU Dortmund offered 7 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was

18.

01. Beschleunigerphysik I / Seminar

Teaching event type	Undergraduate Seminar
Weekly semester hours	1
Languages	German
Semesters	2024-1
Link	https://delta.tu-dortmund.de/studium/lehrveranstaltungen/beschleunigerphysik/#/

02. Beschleunigerphysik I

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://delta.tu-dortmund.de/studium/lehrveranstaltungen/beschleunigerphysik/#/

03. Beschleunigerphysik I / Übung

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-12024-2
Link	https://delta.tu-dortmund.de/studium/lehrveranstaltungen/beschleunigerphysik/#/

04. Beschleunigerphysik II

Teaching event type	Lecture
Weekly semester hours	4
Languages	German
Semesters	2024-2
Link	https://delta.tu-dortmund.de/studium/lehrveranstaltungen/beschleunigerphysik/#/

05. Beschleunigerphysik II / Seminar

Teaching event type	Graduate Seminar
Weekly semester hours	4
Languages	German
Semesters	2024-2
Link	https://delta.tu-dortmund.de/studium/lehrveranstaltungen/beschleunigerphysik/#/

06. Beschleunigerphysik II / Übung

Teaching event type	Lecture Exercise
Weekly semester hours	4
Languages	German
Semesters	2024-12024-2
Link	https://delta.tu-dortmund.de/studium/lehveranstaltungen/beschleunigerphysik/#/

07. Beschleunigerphysik und Synchrotronstrahlung, Anwendungen in der Festkörperphysik

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-12024-2
Link	https://delta.tu-dortmund.de/studium/lehveranstaltungen/beschleunigerphysik-und-synchro...

DOCTORAL THESES

From 2010 onwards, a total of 8 doctoral theses have been published in the field of accelerator research at TU Dortmund. Of these, 8 led to science degrees and none to engineering degrees. 1 were written by women and 7 by men.

01. Gerrit Schünemann (2017)

Development and application of a universal distributed data acquisition system for orbit feedback applications on electron and hadron synchrotrons

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.17877/DE290R-17797
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/dd1a49c2-fc8f-492f-9b03-f8bb0455298e/download
about facility	
in affiliation with	

02. Bernard Riemann (2016)

The Bilinear-Exponential Closed-Orbit Model and its Application to Storage Ring Beam Diagnostics

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
doi	10.17877/DE290R-17221
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/053b7447-893f-40fe-a3c9-53da088763e6/download
about facility	BESSY II
in affiliation with	HZB

03. Peter Ungelenk (2015)

Generation and Detection Schemes for Laser-Induced Coherent Terahertz Radiation at the Electron Storage Ring Delta

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
doi	10.17877/DE290R-17221
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/053b7447-893f-40fe-a3c9-53da088763e6/download
about facility	BESSY II
in affiliation with	HZB

04. Alessandro Ferrarotto (2015)

Entwicklung eines TM110-HF-Deflektors zur longitudinal aufgelösten transversalen Strahldiagnose von Pikosekunden-Elektronenbunchen im MeV-Bereich

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
doi	10.17877/DE290R-16520
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/9af29498-9cc8-4ddc-9de8-0882989d6396/download
about facility	
in affiliation with	

05. Markus Höner (2015)

Investigation of transient processes at the DELTA electron storage ring using a digital bunch-by-bunch feedback system

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
doi	10.17877/DE290R-7517
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/7f9e3b5a-c9bb-44e7-8188-58c5e320d5d8/download
about facility	
in affiliation with	

06. Maryam Huck (2015)

Characterization of VUV pulses from the short-pulse facility at DELTA and steps towards pump-probe experiments

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2015
Language	en
doi	10.17877/DE290R-7053
fulltext link	https://eldorado.tu-dortmund.de/bitstream/2003/33920/1/Dissertation.pdf
about facility	
in affiliation with	

07. Jonathan Fürsch (2014)

Lebensdauerverbesserung und Strahlstabilisierung durch longitudinale Phasenmodulation am Elektronenspeicherring DELTA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	de
doi	10.17877/DE290R-6524
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/102166de-d7d3-4c9b-83e7-39c616c9a348/download
about facility	
in affiliation with	

08. Christian Böhme (2011)

Untersuchungen zur Profilmessung von Hadronenstrahlen mittels Restgaslumineszenz und -ionisation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	de
doi	10.17877/DE290R-13409
fulltext link	https://eldorado.tu-dortmund.de/bitstreams/b6ad8999-84f3-42c1-9e59-044d19b9f9b2/download
about facility	LHC
in affiliation with	CERN

MAX PLANCK INSTITUTE FOR PHYSICS

GENERAL INFORMATION

Short label for the unit	en	MPP
	de	MPP
Official name of the organization	en	Max Planck Institute for Physics
	de	Max-Planck-Institut für Physik
One-line summary	en	Contributes to particle accelerator R&D; partners on CERN projects, operates detector test beams.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Max Planck Institute for Physics		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

DRESDEN UNIVERSITY OF TECHNOLOGY

GENERAL INFORMATION

Short label for the unit	en	TU Dresden
	de	TU Dresden
Official name of the organization	en	Dresden University of Technology
	de	Technische Universität Dresden
One-line summary	en	Drives accelerator research; affiliated with ExciteLab and technology-based high-tech startup accelerators.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Dresden University of Technology		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

From 2010 onwards, a total of 3 doctoral theses have been published in the field of accelerator research at TU Dresden. Of these, 3 led to science degrees and none to engineering degrees. 0 were written by women and 3 by men.

01. Pengnan Lu (2017)

Optimization of an SRF gun for high bunch charge applications at ELBE

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://tud.qucosa.de/api/qucosa%3A30272/attachment/ATT-0/
about facility	
in affiliation with	

02. Hannes Vennekate (2017)

Emittance compensation for SRF photoinjectors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://tud.qucosa.de/api/qucosa%3A30555/attachment/ATT-0/
about facility	
in affiliation with	

03. Martin Greschner (2013)

Aufbau und Inbetriebnahme einer Photoneutronenquelle

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
fulltext link	https://tud.qucosa.de/api/qucosa%3A27060/attachment/ATT-0/
about facility	
in affiliation with	

UNIVERSITY OF BONN

GENERAL INFORMATION

Short label for the unit	en	Uni Bonn
	de	Uni Bonn
Official name of the organization	en	University of Bonn
	de	Rheinische Friedrich-Wilhelms-Universität Bonn
One-line summary	en	Studies particle/accelerator physics; operates ELSA electron stretcher accelerator.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Bonn		-	-	-	-	-	-	-	-	-	-	-	-	0
Physikalisches Institut		-	-	-	-	-	-	-	-	-	-	-	-	0
Group ELSA		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

01. ELSA

Type	en	
	de	
current life cycle	en	3.1 Operation
label	en	ELSA
	de	ELSA
tagline	en	Elektronenstretcher-Anlage
	de	Elektronenstretcher-Anlage
description	en	Die Elektronen-Stretcher-Anlage ELSA wird von der Universität Bonn im Land Nordrhein-Westfalen betrieben. Sie besteht aus drei Stufen (Injektor-LINAC, Booster-Synchrotron und Stretcher-Ring) und liefert einen Strahl polarisierter oder unpolarisierter Elektronen mit variabler Energie von maximal 3,2 GeV. Für das Hauptforschungsgebiet, die Hadronenphysik, können Ströme bis zu 1 nA extrahiert und zu den Experimentierplätzen CB-ELSA und BGOOD transferiert werden.
	de	Die Elektronen-Stretcher-Anlage ELSA wird von der Universität Bonn im Land Nordrhein-Westfalen betrieben. Sie besteht aus drei Stufen (Injektor-LINAC, Booster-Synchrotron und Stretcher-Ring) und liefert einen Strahl polarisierter oder unpolarisierter Elektronen mit variabler Energie von maximal 3,2 GeV. Für das Hauptforschungsgebiet, die Hadronenphysik, können Ströme bis zu 1 nA extrahiert und zu den Experimentierplätzen CB-ELSA und BGOOD transferiert werden.

TEACHING

TEACHING OFFERS

In academic year 2024/25, Uni Bonn offered 2 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 2.

01. Accelerator Physics / Beschleunigerphysik I

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	English
Semesters	2024-1

02. Advanced Accelerator Physics

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-2

DOCTORAL THESES

From 2010 onwards, a total of 12 doctoral theses have been published in the field of accelerator research at Uni Bonn. Of these, 12 led to science degrees and none to engineering degrees. 1 were written by women and 11 by men.

01. Dennis Proft (2018)

Optimierung des Beschleunigerbetriebs für Experimente zur Hadronen- und Detektorphysik an der Elektronen-Stretcher-Anlage ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	de
doi	20.500.11811/8054
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/8054/5534.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

02. Nikolas Heurich (2017)

Die externe Strahlführung für Detektortests X3ed an der Elektronen-Stretcher-Anlage ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	de
doi	20.500.11811/7310
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/7310/4919.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

03. Jan Felix Schmidt (2017)

Spindynamik in Elektronensynchrotronen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	de
doi	20.500.11811/7256
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/7256/4831.pdf?sequence=1&isAllowed=y
about facility	
in affiliation with	

04. Manuel Schedler (2015)

Intensitäts- und Energieerhöhung an ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
doi	20.500.11811/6574
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/6574/4207.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

05. Oliver Boldt (2014)

Spindynamik von Elektronenstrahlen in Kreisbeschleunigern

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	de
doi	20.500.11811/6110
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/6110/3645.pdf?sequence=1&isAllowed=y
about facility	
in affiliation with	

06. Dominik Heiliger (2014)

Erzeugung intensiver spinpolarisierter Elektronenstrahlen an der Beschleunigeranlage ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	de
doi	20.500.11811/6159
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/6159/3725.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

07. Sven Zander (2013)

Optische Strahldiagnose an der Elektronen-Stretcher-Anlage ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
doi	20.500.11811/5770
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/5770/3365.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

08. Thorsten R. Pusch (2012)

Bestimmung von Intensität und Position des extrahierten Elektronenstrahls an ELSA mittels Hochfrequenzresonatoren

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	de
doi	20.500.11811/5301
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/5301/2824.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

09. André Roth (2012)

Breitbandige Feedback-Systeme zur Dämpfung kohärenter Strahlinstabilitäten am Stretcherring ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	de
doi	20.500.11811/5603
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/5603/3090.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

10. Fabian Klarner (2011)

Konzeption, Aufbau und Inbetriebnahme eines neuen Vorbeschleunigersystems an ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	de
doi	20.500.11811/5077
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/5077/2721.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

11. Dominic Markus Welsch (2010)

Investigation and optimization of transverse non-linear beam dynamics in the high-energy storage ring HESR

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2010
Language	en
doi	20.500.11811/4554
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/4554/2082.pdf?sequence=1&isAllowed=y
about facility	
in affiliation with	FAIR

12. Maren Eberhardt (2010)

Messung und Korrektur der Arbeitspunkte während der Energierampe am Stretcherring von ELSA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2010
Language	de
doi	20.500.11811/4701
fulltext link	https://bonndoc.ulb.uni-bonn.de/xmlui/bitstream/handle/20.500.11811/4701/2378.pdf?sequence=1&isAllowed=y
about facility	ELSA
in affiliation with	

TECHNICAL UNIVERSITY OF DARMSTADT

GENERAL INFORMATION

Short label for the unit	en	TU Darmstadt
	de	TU Darmstadt
Official name of the organization	en	Technical University of Darmstadt
	de	Technische Universität Darmstadt
One-line summary	en	Advances superconducting electron accelerators; operates S-DALINAC.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Technical University of Darmstadt		9	15	11	73	11	5	93	31	-	26	98	-	12
Department of Electrical Engineering and Information Technology		5	6	-	32	-	-	12	31	-	11	32	-	43
Institute for Accelerator Science und Electromagnetic Fields		5	6	-	32	-	-	12	31	-	11	32	-	43
Chair of Electromagnetic Field Theory		1	2	-	6	-	-	9	-	-	3	6	-	9
Group: ACHIP		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: FAIR		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: PUMA		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Laser Ion Generation, Handling and Transport		-	-	-	-	-	-	-	-	-	-	-	-	0
Chair of Electromagnetic Field Theory		1	-	-	2	-	-	3	-	-	-	3	-	3
Chair of Electromagnetic Field Theory		3	4	-	24	-	-	-	31	-	8	23	-	31
Group: Computational Magnet Analysis and Dynamics		-	1	-	5	-	-	-	6	-	3	3	-	6
Group: Coupled Problems		-	1	-	-	-	-	-	1	-	-	1	-	1
Group: High Frequency Technology		-	1	-	-	-	-	-	1	-	-	1	-	1
Group: Beam Dynamics		-	1	-	-	-	-	-	1	-	1	-	-	1
Department of Physics		4	9	11	41	11	5	81	-	-	15	66	-	81
Institute for Nuclear Physics		4	9	11	41	11	5	81	-	-	15	66	-	81
Group Enders		1	-	2	3	2	3	11	-	-	-	11	-	11
Group Pietralla		1	8	4	20	5	-	38	-	-	10	28	-	38

NAME	MEMBERS	SSOR	HER	DOC	DEUT	DEUT	DEUT	ICIST	VEER	HER	MALE	MALE	MARY	SUM
Group Obertelli		1	-	4	7	2	-	14	-	-	4	10	-	14
Group Roth		1	1	1	11	2	2	18	-	-	1	17	-	18

FACILITIES

USER FACILITIES

TEST FACILITIES

01. S-DALINAC

Type	en	Energy Recovery Linac
	de	Linearbeschleuniger mit Energierückgewinnung
current life cycle	en	3.1 Operation
label	en	S-DALINAC
	de	S-DALINAC
tagline	en	Superconducting-Darmstadt-LINear-ACcelerator
	de	Supraleitender Linearbeschleuniger in Darmstadt
description	en	Supraleitender Elektronenbeschleuniger, der sowohl im Dauerstrich- als auch im Energierückgewinnungs-Modus betrieben werden kann.
	de	Supraleitender Elektronenbeschleuniger, der sowohl im Dauerstrich- als auch im Energierückgewinnungs-Modus betrieben werden kann.

TEACHING

TEACHING OFFERS

In academic year 2024/25, TU Darmstadt offered 13 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 29.

01. Accelerator Physics

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

02. Particle Accelerator Technology

Teaching event type	Undergraduate Seminar
Weekly semester hours	4
Languages	German
Semesters	2024-1
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

03. Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

04. Physik dichter Plasmen mit Schwerionen- und Laserstrahlen

Teaching event type	Seminar
Weekly semester hours	2
Languages	GermanEnglish
Semesters	2024-1
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

05. Radio Frequency Systems for Particle Accelerators

Teaching event type	Lecture
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

06. Radio Frequency Systems for Particle Accelerators

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

07. Accelerator Physics and Technology

Teaching event type	Seminar
Weekly semester hours	1
Languages	GermanEnglish
Semesters	2024-1
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

08. Projektoberseminar Beschleunigertechnik

Teaching event type	Undergraduate Seminar
Weekly semester hours	4
Languages	German
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

09. Projektseminar Beschleunigertechnik

Teaching event type	Undergraduate Seminar
Weekly semester hours	4
Languages	German
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

10. Simulation von Strahldynamik und elektromagnetischen Feldern in Teilchenbeschleunigern

Teaching event type	Lecture
Weekly semester hours	2
Languages	EnglishEnglish
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

11. Röntgenlicht-Freie-Elektronen-Laser

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-2
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

12. Laser particle acceleration and secondary radiation sources (Tutorial)

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	GermanEnglish
Semesters	2024-1

13. Numerical methods of accelerator physics

Teaching event type	Lecture
Weekly semester hours	2
Languages	GermanEnglish
Semesters	2024-1
Link	https://www.tucan.tu-darmstadt.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

DOCTORAL THESES

From 2010 onwards, a total of 81 doctoral theses have been published in the field of accelerator research at TU Darmstadt. Of these, 66 led to science degrees and 15 to engineering degrees. 11 were written by women and 70 by men.

01. Maxim Saifulin (2024)

Fast scintillating ZnO ceramics for relativistic heavy-ion beam diagnostics

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.26083/tuprints-00026525
fulltext link	https://tuprints.ulb.tu-darmstadt.de/26525/1/Saifulin_PhD-Thesis_final.pdf
about facility	UNILAC
in affiliation with	FAIR

02. Haress Nazary (2024)

Towards Ion Stopping Power Experiments with the Laser-Driven LIGHT Beamline

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.26083/tuprints-00027874
fulltext link	https://tuprints.ulb.tu-darmstadt.de/27874/1/HNazary_Dissertation.pdf
about facility	
in affiliation with	

03. Oliver Papst (2024)

Exploration of nuclear-structure effects on averaged decay quantities in the quasicontinuum

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.26083/tuprints-00011437
fulltext link	https://tuprints.ulb.tu-darmstadt.de/11437/1/2024_Oliver_Papst_Dissertation.pdf
about facility	S-DALINAC
in affiliation with	

04. Manuel Dutine (2024)

Entwicklung und test einer strahlpositionsmessung für den mehrfach rezirkulierenden ERL-betrieb des S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.26083/tuprints-00027838
fulltext link	https://tuprints.ulb.tu-darmstadt.de/27838/1/Dissertation_Dutine_Veröffentlichung.pdf
about facility	S-DALINAC
in affiliation with	

05. Pascal Simon (2023)

Carbon-Based Materials for High-Power Accelerator Components Exposed to Extreme Radiation Conditions

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00024606
fulltext link	https://tuprints.ulb.tu-darmstadt.de/24606/1/20231013_Dissertation_PSimon.pdf
about facility	UNILAC
in affiliation with	FAIR

06. Verena Velthaus (2023)

Mitigation of heavy ion-stimulated gas desorption from accelerator components

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2023
Language	en
doi	10.26083/tuprints-00024344
fulltext link	https://tuprints.ulb.tu-darmstadt.de/24344/1/Diss_Velthaus_für_Veröffentlichung.pdf
about facility	UNILAC
in affiliation with	FAIR

07. Jan-Paul Alexander Hucka (2023)

Control System for the Next Generation In-flight Separator Super-FRS applied to New Isotope Search with the FRS.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00023299
about facility	Super-FRS
in affiliation with	FAIR

08. Martin Metternich (2023)

Shaping of laser-accelerated ion beams for applications in high energy density physics

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00024395
fulltext link	https://repository.gsi.de/record/341044/files/phd_mmetternich.pdf
about facility	
in affiliation with	

09. Yannik Zobus (2023)

Design and implementation of a high-contrast, millijoule-level ultrafast optical parametric amplifier for high-intensity lasers

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00024200
fulltext link	https://tuprints.ulb.tu-darmstadt.de/24200/1/dissertation_yzobus_finalpublication.pdf
about facility	
in affiliation with	

10. Gerhart Steinhilber (2023)

The electron-gamma coincidence set-up at the S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00022993
fulltext link	https://tuprints.ulb.tu-darmstadt.de/22993/1/GSteinhilber_Dissertation.pdf
about facility	S-DALINAC
in affiliation with	

11. Elif Sahin (2023)

Lifetime measurement of the yrast 2^+ state of ^{90}W

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00024403
fulltext link	https://tuprints.ulb.tu-darmstadt.de/24403/9/Dissertation_Sahin_v2.pdf
about facility	
in affiliation with	

12. Tobias Eggert (2023)

Development of a Cryogenic DC Photoelectron Gun for GaAs Cathodes with Increased Charge Lifetime

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.26083/tuprints-00023275
fulltext link	https://tuprints.ulb.tu-darmstadt.de/23275/1/Dissertation_Tobias_Eggert.pdf
about facility	
in affiliation with	

13. Dmytro Dmytriiev (2022)

Design of a position sensitive resonant schottky detector for the rare-RI ring in RIKEN

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.11588/HEIDOK.00031183
fulltext link	https://archiv.ub.uni-heidelberg.de/volltextserver/31183/1/Dima_thesis_2021.pdf
about facility	CRS-DALINAC
in affiliation with	FAIR

14. Konstantin Thilo Mohr (2022)

First laser spectroscopy of Mg⁺ at CRYRING@ESR and He-like boron at HITRAP

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.26083/TUPRINTS-00022451
fulltext link	https://tuprints.ulb.tu-darmstadt.de/22451/1/Dissertation_KonstantinMohr_v1.pdf
about facility	
in affiliation with	FAIR

15. Sergei Glukhov (2022)

Multipass beam breakup instability countermeasures in energy recovery linacs

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.26083/tuprints-00022915
fulltext link	https://tuprints.ulb.tu-darmstadt.de/22915/1/2022-06-03_Glukhov_Sergei.pdf
about facility	
in affiliation with	

16. Tim Wolz (2022)

Controlled manipulation of atoms in rydberg quantum states for application in experiments with antihydrogen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.26083/tuprints-00020310
fulltext link	https://tuprints.ulb.tu-darmstadt.de/20310/1/Dissertation_WOLZ_online.pdf
about facility	
in affiliation with	

17. Simon Weih (2022)

Injector Optimization at the Superconducting Darmstadt Linear Electron Accelerator S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.26083/tuprints-00020632
fulltext link	https://tuprints.ulb.tu-darmstadt.de/20632/1/Dissertation_Weih_genehmigt_v1.pdf
about facility	S-DALINAC
in affiliation with	

18. Tobias Beck (2021)

Novel decay properties of positive-parity low-spin states of deformed gadolinium and dysprosium nuclei

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.26083/tuprints-00019255
fulltext link	https://tuprints.ulb.tu-darmstadt.de/19255/1/Beck_PhD_2021_online.pdf
about facility	
in affiliation with	

19. Ralph Kern (2021)

Isovector one-quadrupole-phonon excitations of heavy vibrational nuclei

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.26083/tuprints-00018581
fulltext link	https://tuprints.ulb.tu-darmstadt.de/18581/20/PhD-RK-TUDa_pub.pdf
about facility	
in affiliation with	

20. Antonio D'Alessio (2020)

B(E2)-stärke des 2+1-zustandes von ^{12}C aus (e,e γ)-daten & neuplanung des vieldraht-driftkammer detektors am QCLAM-spektrometer

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00014222
fulltext link	https://tuprints.ulb.tu-darmstadt.de/14222/1/Dissertation_DAlessio.pdf
about facility	
in affiliation with	

21. Udo Friman-Gayer (2020)

Probing nuclear structure relevant for neutrinoless double-beta decay with nuclear resonance fluorescence

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00011385
fulltext link	https://tuprints.ulb.tu-darmstadt.de/11385/7/dissertation_ug.pdf
about facility	
in affiliation with	

22. Johannes Wiederhold (2020)

γ -spectroscopic determination of mean lifetimes and magnetic moments of excited nuclear states

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00011446
fulltext link	https://tuprints.ulb.tu-darmstadt.de/11446/1/dissertation_jwiederhold_14022020.pdf
about facility	
in affiliation with	

23. Philipp Ries (2020)

The low-lying photonresponse of the even-even chromium isotopes

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00011909
fulltext link	https://tuprints.ulb.tu-darmstadt.de/11909/13/Diss_PRies_publication.pdf
about facility	
in affiliation with	

24. Tobias Klaus (2020)

Elektrische dipol-polarisierbarkeit und spin-M1-stärke aus 90^ozr(p,p^o)-daten unter 0?

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00012687
fulltext link	https://tuprints.ulb.tu-darmstadt.de/12687/1/Dissertation_TobiasKlaus_Final_2020.pdf
about facility	
in affiliation with	

25. Michaela Hilcker (2020)

A superfluid liquid helium target for electron scattering experiments at the S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2020
Language	en
urn	urn:nbn:de:tuda-tuprints-12811
fulltext link	https://tuprints.ulb.tu-darmstadt.de/12811/1/20200714%20-%20A%20superfluid%20liquid%20Helium%20Target%20for%20Electron%20Scattering%20ExpDALINAC.pdf
about facility	S-DALINAC
in affiliation with	

26. Marcel Berger (2020)

Kernstrukturuntersuchungen bis zur teilchenseparationsschwelle mit der methode der kernresonanzfluoreszenz

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	10.25534/tuprints-00012852
fulltext link	https://tuprints.ulb.tu-darmstadt.de/19255/1/Beck_PhD_2021_online.pdf
about facility	
in affiliation with	

27. Pavlos Koseoglou (2019)

Lifetime measurements in the neutron-rich ^{148}Ce nuclide at the low-Z boundary of the $N=90$ shape-phase transition

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
urn	urn:nbn:de:tuda-tuprints-85923
fulltext link	https://tuprints.ulb.tu-darmstadt.de/8592/15/Phd_thesis_Koseoglou.pdf
about facility	
in affiliation with	

28. Waldemar Witt (2019)

Type-II shell evolution in ^{98}Zr

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
urn	urn:nbn:de:tuda-tuprints-85367
about facility	
in affiliation with	

29. Philipp Napiralla (2019)

Employing γ -ray tracking as an event-discrimination technique for γ -spectroscopy with AGATA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
doi	10.25534/tuprints-00009673
fulltext link	https://tuprints.ulb.tu-darmstadt.de/9673/1/Dissertation_Napiralla.pdf
about facility	
in affiliation with	

30. Dmitry Bazyl (2019)

Development of an SRF reduced-beta cavity for the injector of the s-dalinac

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2019
Language	en
urn	urn:nbn:de:tuda-tuprints-87910
fulltext link	https://tuprints.ulb.tu-darmstadt.de/8791/1/2019-05-20_Bazyl_Dmitry.pdf
about facility	S-DALINAC
in affiliation with	

31. Andrea Passarelli (2019)

Characterization of impedance and wakefields of accelerator devices in the short bunch limit and beam dynamics effects for the CLIC Damping Rings

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2019
Language	en
urn	urn:nbn:de:tuda-tuprints-86441
fulltext link	https://tuprints.ulb.tu-darmstadt.de/8644/1/2019-04-23_Passarelli_Andrea.pdf
about facility	
in affiliation with	

32. Michael Reese (2018)

Intermediate-energy coulomb excitation with the PreSPEC-AGATA setup

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-73279
about facility	
in affiliation with	FAIR

33. Marc Lettmann (2018)

Spectroscopy of neutron-rich germanium and krypton isotopes - a quest of triaxiality

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-73821
about facility	
in affiliation with	

34. Cesar Lizarazo (2018)

Gamma-ray spectroscopy of $^{92,94}\text{Se}$ isomeric decay

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-72660
about facility	
in affiliation with	

35. Lewin Eidam (2018)

Laser cooling of intense relativistic ion beams

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-71523
fulltext link	https://tuprints.ulb.tu-darmstadt.de/7152/1/Thesis_eidam.pdf
about facility	
in affiliation with	

36. Annalisa Romano (2018)

Electron cloud formation in CERN particle accelerators and its impact on the beam dynamics

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-80716
fulltext link	https://tuprints.ulb.tu-darmstadt.de/8071/1/2018-08-20_Romano_Annalisa.pdf
about facility	
in affiliation with	

37. Yaoshuo Yuan (2018)

Space-charge driven transverse beam instabilities in synchrotrons

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
urn	urn:nbn:de:tuda-tuprints-82155
fulltext link	https://tuprints.ulb.tu-darmstadt.de/8215/1/2018-11-06_Yuan_Yaoshuo.pdf
about facility	
in affiliation with	

38. Oliver Bitterling (2017)

Development of Radiation-Tolerant Components for the Quench Detection System at the CERN Large Hadron Collider

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.26083/tuprints-00017594
fulltext link	https://tuprints.ulb.tu-darmstadt.de/6300/1/PhDThesis_V2.pdf
about facility	LHC
in affiliation with	

39. Piotr Szwangruber (2017)

Quench study for FAIR magnets

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2017
Language	en
urn	urn:nbn:de:tuda-tuprints-72079
fulltext link	https://tuprints.ulb.tu-darmstadt.de/7207/1/PhD_Piotr_Szwangruber_Quench_Study_for_FAIR_Magnets.pdf
about facility	
in affiliation with	FAIR

40. Robert Stegmann (2017)

One-quadrupole-phonon states of heavy vibrational nuclei studied in coulomb excitation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
urn	urn:nbn:de:tuda-tuprints-69399
about facility	
in affiliation with	

41. Neeraj Kurichiyanil (2017)

Design and construction of a test stand for photocathode research and experiments.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5903/8/Dissertation_print2.pdf
about facility	
in affiliation with	

42. Michaela Arnold (2017)

Auslegung, Planung und Aufbau einer dritten Rezirkulation mit ERL-Modus für den S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2017
Language	de
urn	urn:nbn:de:tuda-tuprints-61940
fulltext link	https://tuprints.ulb.tu-darmstadt.de/6194/1/Dissertation_Michaela_Arnold_oL_ULB.pdf
about facility	S-DALINAC
in affiliation with	

43. Thore Bahlo (2017)

Entwurf eines Møllerpolarimeters und Entwicklung einer aktiven Phasenstabilisierung für den Injektor des S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	de
fulltext link	https://tuprints.ulb.tu-darmstadt.de/6849/7/dissertation_veroeffentlicht.pdf
about facility	S-DALINAC
in affiliation with	

44. Jacopo Corno (2017)

Numerical methods for the estimation of the impact of geometric uncertainties on the performance of electromagnetic devices

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2017
Language	en
urn	urn:nbn:de:tuda-tuprints-70389
fulltext link	https://tuprints.ulb.tu-darmstadt.de/7038/7/thesis.pdf
about facility	
in affiliation with	

45. Yves Hackl (2017)

Effiziente Simulation magnetischer Bauteile

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2017
Language	de
urn	urn:nbn:de:tuda-tuprints-58717
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5871/1/V02—Einreichen_Online_21-02-2017.pdf
about facility	
in affiliation with	

46. Ivan Karpov (2017)

Damping of coherent oscillations in intense ion beams

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2017
Language	en
urn	urn:nbn:de:tuda-tuprints-58717
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5871/1/V02—Einreichen_Online_21-02-2017.pdf
about facility	
in affiliation with	

47. Thomas Schösser (2017)

Entwicklung und Implementierung einer EPICS-basierten Beschleuniger-Steuerungsschnittstelle und automatisierten Strahloptimierung am S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	de
urn	urn:nbn:de:tuda-tuprints-65181
fulltext link	https://tuprints.ulb.tu-darmstadt.de/6518/1/Dissertation_Thomas_Schösser.pdf
about facility	S-DALINAC
in affiliation with	

48. Martha Liliana Cortes Sua (2016)

Inelastic scattering of Ni and Zn isotopes off a proton target

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-55953
about facility	
in affiliation with	FAIR

49. Markus Zweidinger (2016)

Kernstrukturuntersuchungen in mittelschweren atomkernen mit der methode der kernresonanzfluoreszenz

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-56154
about facility	
in affiliation with	

50. Christoph Kremer (2016)

Linking partial and quasi dynamical symmetries in rotational nuclei and shell evolution in 96-Zr

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-52687
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5268/1/Dissertation_published.pdf
about facility	
in affiliation with	

51. Jonny Hubertus Birkhan (2016)

Elektrische dipol-polarisierbarkeit und spin-M1-stärke aus $^{48}\text{Ca}(p,p')$ -daten unter 0°

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-52167
about facility	
in affiliation with	

52. Martin Espig (2016)

Entwicklung, Aufbau und Charakterisierung einer variabel repetierenden, spinpolarisierten Elektronenkanone mit invertierter Isolatorgeometrie

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	de
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5328/1/Dissertation_Espig_Martin.pdf
about facility	
in affiliation with	

53. Oliver Köster (2016)

Combining rotating-coil measurements of large-aperture accelerator magnets.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-56970
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5697/1/PhD_final.pdf
about facility	
in affiliation with	

54. Uwe Niedermayer (2016)

Determination of beam coupling impedance in the frequency domain

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-51573
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5157/1/main%20-%20Kopie.pdf
about facility	
in affiliation with	

55. Peter Schmidt (2016)

Radiation pressure acceleration and laser induced focusing

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2016
Language	en
urn	urn:nbn:de:tuda-tuprints-56659
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5665/1/final_PhD_Thesis.pdf
about facility	
in affiliation with	

56. Damian Ralet (2015)

Lifetime measurement in the mass region A<code>126</code>100 with the PreSPEC-AGATA setup

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
urn	urn:nbn:de:tuda-tuprints-46132
fulltext link	https://tuprints.ulb.tu-darmstadt.de/4613/13/TUDthesis.pdf
about facility	
in affiliation with	FAIR

57. Christian Stahl (2015)

New methods for the γ -ray spectroscopy with position-sensitive detector systems

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
urn	urn:nbn:de:tuda-tuprints-49556
about facility	
in affiliation with	

58. Christopher Romig (2015)

Investigation of nuclear structure with relative self-absorption measurements

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
isbn	978-3-8439-2021-6
urn	urn:nbn:de:tuda-tuprints-44468
fulltext link	https://tuprints.ulb.tu-darmstadt.de/4446/1/doctoral-thesis-romig-final.pdf
about facility	
in affiliation with	

59. Paul A. G6rgen (2015)

Transverse beam transfer functions of relativistic ion bunches

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	en
urn	urn:nbn:de:tuda-tuprints-50230
fulltext link	https://tuprints.ulb.tu-darmstadt.de/5023/2/Thesis.pdf
about facility	
in affiliation with	

60. Giulia Guastalla (2014)

Shell structure in the vicinity of the doubly magic 100Sn via coulomb excitation at PreSPEC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2014
Language	en
urn	urn:nbn:de:tuda-tuprints-42853
about facility	
in affiliation with	FAIR

61. Timo Milosic (2014)

Feasibility Study on Longitudinal Phase-Space Measurements at GSI UNILAC using Charged-Particle Detectors

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
fulltext link	https://tuprints.ulb.tu-darmstadt.de/4321/1/Milosic_PhD-Thesis_2013.pdf
about facility	UNILAC
in affiliation with	FAIR

62. Jacob Daniel Beller (2014)

Systematische untersuchung exotischer zerfallskanäle der scherenmode in gadoliniumisotopen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
urn	urn:nbn:de:tuda-tuprints-38590
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3859/7/thesis.pdf
about facility	
in affiliation with	

63. Christopher Walz (2014)

The two-photon decay of the 11/2- isomer of ^{137}Ba and mixed-symmetry states of $^{92,94}\text{Zr}$ and ^{94}Mo

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
urn	urn:nbn:de:tuda-tuprints-40566
about facility	
in affiliation with	

64. Thomas Möller (2014)

Aspects of nuclear collectivity studied in projectile coulomb excitation experiments

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
urn	urn:nbn:de:tuda-tuprints-40263
fulltext link	https://tuprints.ulb.tu-darmstadt.de/4026/2/Thesis.pdf
about facility	
in affiliation with	

65. Andreas Krugmann (2014)

Off-yrast low-spin structure of deformed nuclei at mass number $a \approx 150$

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2014
Language	en
urn	urn:nbn:de:tuda-tuprints-42898
fulltext link	https://tuprints.ulb.tu-darmstadt.de/4289/13/diss_krugmann_ulb_03.pdf
about facility	
in affiliation with	

66. Abdulrahman Scheikh Obeid (2013)

Excitation strengths and transition radii differences of one-phonon quadrupole excitations from electron scattering on $^{92,94}\text{Zr}$ and ^{94}Mo

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	en
urn	urn:nbn:de:tuda-tuprints-37742
about facility	
in affiliation with	

67. Christopher Bauer (2013)

Level lifetimes and quadrupole moments from projectile coulomb excitation of $A < 126 > 130$ nuclei

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	en
isbn	978-3-8439-1124-5
about facility	
in affiliation with	

68. Thorsten Kürzeder (2013)

Entwicklung, Aufbau und Test eines neuen Kryostatmoduls für den S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
urn	urn:nbn:de:tuda-tuprints-33533
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3353/1/kuerzederdissonline.pdf
about facility	S-DALINAC
in affiliation with	

69. Zsolt Léczi (2013)

Laser ion acceleration from a double-layer metal foil.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2013
Language	en
urn	urn:nbn:de:tuda-tuprints-33359
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3335/1/PHD_final.pdf
about facility	
in affiliation with	

70. Fedor Petrov (2013)

Electron clouds in high energy hadron accelerators.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2013
Language	en
urn	urn:nbn:de:tuda-tuprints-35893
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3589/7/TUDthesis_Petrov.pdf
about facility	
in affiliation with	

71. Sven Thorsten Sievers (2013)

Untersuchung von Piezoaktoren zur Verbesserung der Frequenzabstimmung und Aufbau eines Quenchortungssystems für die Beschleunigungsstrukturen des S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
urn	urn:nbn:de:tuda-tuprints-35373
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3537/1/Dissertation%20Sven%20Sievers%20onlineversion.pdf
about facility	S-DALINAC
in affiliation with	

72. Markus Wagner (2013)

Erzeugung und Untersuchung gepulster polarisierter Elektronenstrahlen am S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
urn	urn:nbn:de:tuda-tuprints-35851
fulltext link	https://tuprints.ulb.tu-darmstadt.de/3585/1/Meine%20Dissertation2.pdf
about facility	S-DALINAC
in affiliation with	

73. Linda Schnorrenberger (2012)

Characterization of gamma-ray detectors with tagged photons

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2012
Language	en
isbn	978-3-8440-0802-9
about facility	
in affiliation with	

74. Babak Alikhani (2012)

Aufbau und Inbetriebnahme des DAGATA-Polarimeters

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	de
urn	urn:nbn:de:tuda-tuprints-29934
about facility	
in affiliation with	

75. Sabrina Appel (2011)

Simulation und Messung longitudinaler Raumladungseffekte in intensiven Ionenstrahlen im SIS18 Synchrotron.

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2011
Language	de
about facility	SIS18
in affiliation with	FAIR

76. Namita Goel (2011)

Spatial characterisation of a 36-fold segmented AGATA detector via a novel scanning system

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2011
Language	en
about facility	
in affiliation with	

77. Matthias Fritzsche (2011)

Untersuchung der pygmy dipol resonanz in 60ni

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	en
isbn	978-3-8439-0198-7
about facility	
in affiliation with	

78. Yuliya Fritzsche (2011)

Aufbau und Inbetriebnahme einer Quelle polarisierter Elektronen am supraleitenden Darmstädter Elektronenlinearbeschleuniger S-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2011
Language	de
urn	urn:nbn:de:tuda-tuprints-28015
fulltext link	https://tuprints.ulb.tu-darmstadt.de/2801/1/Fritzsche-Dissertation.pdf
about facility	S-DALINAC
in affiliation with	

79. Laurent Coquard (2010)

Evolution of the one-quadrupole phonon $2+_{1,ms}$ mixed-symmetry state in $^{124,126,128,130,132}\text{Xe}$

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2010
Language	en
urn	urn:nbn:de:tuda-tuprints-22116
about facility	
in affiliation with	

80. Stefan Paret (2010)

Transverse Schottky spectra and beam transfer functions of coasting ion beams with space charge

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2010
Language	en
urn	urn:nbn:de:tuda-tuprints-21347
fulltext link	https://tuprints.ulb.tu-darmstadt.de/2134/1/thesisStefanParet.pdf
about facility	
in affiliation with	

81. Sebastian Hermann Müller (2009)

Bestimmung von photo-neutron-wirkungsquerschnitten in kernen der seltenen erden für den astrophysikalischen p-prozess mit bremsstrahlung am s-DALINAC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2009
Language	en
isbn	978-3-86853-083-4
about facility	S-DALINAC
in affiliation with	

RWTH AACHEN UNIVERSITY

GENERAL INFORMATION

Short label for the unit	en	RWTH Aachen
	de	RWTH Aachen
Official name of the organization	en	RWTH Aachen University
	de	Rheinisch-Westfälische Technische Hochschule Aachen
One-line summary	en	Innovates accelerator physics; hosts labs for beam and detector R&D, linked with Innovation Factory ecosystem.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
RWTH Aachen University		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

In academic year 2024/25, RWTH Aachen offered 3 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 5.

01. Accelerator Physics Seminar

Teaching event type	Graduate Seminar
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://online.rwth-aachen.de/RWTHonline/ee/ui/ca2/app/desktop/#/slc.tm.cp/student/cours...

02. Introduction to Accelerator Physics (Lecture)

Teaching event type	Lecture
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://online.rwth-aachen.de/RWTHonline/ee/ui/ca2/app/desktop/#/slc.tm.cp/student/cours...

03. Introduction to Accelerator Physics (Exercises)

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	English
Semesters	2024-1
Link	https://online.rwth-aachen.de/RWTHonline/ee/ui/ca2/app/desktop/#/slc.tm.cp/student/cours...

DOCTORAL THESES

From 2010 onwards, a total of 8 doctoral theses have been published in the field of accelerator research at RWTH Aachen. Of these, 8 led to science degrees and none to engineering degrees. 1 were written by women and 7 by men.

01. Jan Henry Hetzel (2018)

Beam-dynamics calculations including magnetic field measurements for the high-energy storage ring (HESR) at FAIR

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.18154/RWTH-2018-223914
fulltext link	https://publications.rwth-aachen.de/record/723015/files/723015.pdf
about facility	
in affiliation with	FAIR

02. Jamal S. Slim (2018)

A novel waveguide RF Wien filter for electric dipole moment measurements of deuterons and protons at the COoler SYnchrotron (COSY)/Jülich

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.18154/RWTH-2018-223914
fulltext link	https://publications.rwth-aachen.de/record/723015/files/723015.pdf
about facility	COSY
in affiliation with	FAIR

03. Nils Hempelmann (2018)

Polarization measurement and manipulation for electric dipole moment measurements in storage rings

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.18154/RWTH-2018-221496
fulltext link	https://publications.rwth-aachen.de/record/718035/files/718035.pdf
about facility	
in affiliation with	

04. Fabian Hinder (2017)

Development of beam diagnostic systems for electric dipole moment measurements at particle accelerators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.18154/RWTH-2018-222658
fulltext link	https://publications.rwth-aachen.de/record/720190/files/720190.pdf
about facility	
in affiliation with	

05. Stanislav Chekmenev (2017)

Investigation of possibilities to measure the deuteron electric dipole moment at storage rings

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.18154/RWTH-2017-03791
fulltext link	https://publications.rwth-aachen.de/record/688413/files/688413.pdf
about facility	
in affiliation with	

06. Fabian Trinkel (2017)

Development of a Rogowski coil beam position monitor for electric dipole moment measurements at storage rings

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.18154/RWTH-2018-00258
fulltext link	https://publications.rwth-aachen.de/record/712121/files/712121.pdf
about facility	
in affiliation with	

07. Marcel Rosenthal (2016)

Experimental benchmarking of spin tracking algorithms for electric dipole moment searches at the cooler synchrotron COSY

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	https://publications.rwth-aachen.de/record/671012/files/671012.pdf
about facility	COSY
in affiliation with	

08. Michaela Schaumann (2015)

Heavy-Ion Performance of the LHC and Future Colliders

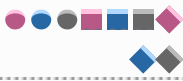
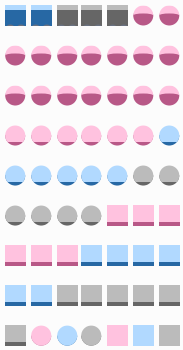
Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2015
Language	en
fulltext link	https://cds.cern.ch/record/2065692/files/CERN-THESIS-2015-195.pdf?version=1
about facility	LHC
in affiliation with	CERN

FRIEDRICH-ALEXANDER-UNIVERSITY ERLANGEN-NÜRNBERG

GENERAL INFORMATION

Short label for the unit	en	Uni Erlangen
	de	Uni Erlangen
Official name of the organization	en	Friedrich-Alexander-University Erlangen-Nürnberg
	de	Friedrich-Alexander-Universität Erlangen-Nürnberg
One-line summary	en	Engaged in accelerator applications; operates electron and ion beam laboratories for research and teaching.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Friedrich-Alexander-University Erlangen-Nürnberg		3	18	16	-	36	6	46	30	3	39	20	-	79
Department of Physics		3	18	16	-	36	6	46	30	3	39	20	-	79
Chair of Laser Physics		3	18	16	-	36	6	46	30	3	39	20	-	79
Group: Accelerator on a Chip International Program		3	9	16	-	36	6	43	27	-	36	17	-	70

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HEINRICH HEINE UNIVERSITY DÜSSELDORF

GENERAL INFORMATION

Short label for the unit	en	HHU
	de	HHU
Official name of the organization	en	Heinrich Heine University Düsseldorf
	de	Heinrich-Heine-Universität Düsseldorf
One-line summary	en	<to be set>
	de	<to be translated>

RESEARCH GROUPS

RESEARCH GROUPS															
NAME	MEMBERS														
	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM		
Heinrich Heine University Düsseldorf	6	-	36	18	81	-	47	47	47	48	48	-	14		
Faculty of Mathematics and Natural Sciences consists	6	-	36	18	81	-	47	47	47	48	48	-	14		
Physics Department	6	-	36	18	81	-	47	47	47	48	48	-	14		
Institute for Laser and Plasma Physics	6	-	36	18	81	-	47	47	47	48	48	-	14		
Group Hidding	6	-	36	18	81	-	47	47	47	48	48	-	14		
													</		

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

HAMBURG UNIVERSITY

GENERAL INFORMATION

Short label for the unit	en	Uni Hamburg
	de	Uni Hamburg
Official name of the organization	en	Hamburg University
	de	Universität Hamburg
One-line summary	en	Engages in accelerator R&D; participates in joint facilities with DESY and European XFEL.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Hamburg University		2	7	-	-	-	-	9	-	-	-	9	-	9
MIN Faculty		2	7	-	-	-	-	9	-	-	-	9	-	9
Department of Physics		2	7	-	-	-	-	9	-	-	-	9	-	9
Institute of Experimental Physics		2	7	-	-	-	-	9	-	-	-	9	-	9
Group Hillert	●	1	2	-	-	-	-	3	-	-	-	3	-	3
Team: Advanced Seeding Techniques for XUV and Angstrom Radiation	●●	-	2	-	-	-	-	2	-	-	-	2	-	2
Team: Superconducting Radio-Frequency Cavities		-	-	-	-	-	-	-	-	-	-	-	-	0
Group Grüner	●	1	5	-	-	-	-	6	-	-	-	6	-	6
Team: Novel X-ray light sources based on plasma-wakefield acceleration	●●●●●	-	5	-	-	-	-	5	-	-	-	5	-	5

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

In academic year 2024/25, Uni Hamburg offered 9 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 22.

01. Accelerator Physics I

Teaching event type	Lecture
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

02. Computer exercises: Accelerator Physics I

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

03. Accelerator Physics II

Teaching event type	Lecture
Weekly semester hours	4
Languages	English
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

04. Computer exercises: Accelerator Physics II

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

05. Exkursion: Accelerator Physics II

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	English
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

06. Promsenimar: Physik und Anwendungen von Laser-Plasma-Beschleunigern: Von medizinischer Bildgebung bis Hochenergiephysik

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

07. Physik und Anwendungen von Laser-Plasma-Beschleunigern: Von medizinischer Bildgebung bis Hochenergiephysik

Teaching event type	Lecture
Weekly semester hours	4
Languages	German
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

08. Proseminar: Theorie und Praxis moderner Beschleuniger

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	English
Semesters	2024-1
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

09. Übungen: Physik und Anwendungen von Laser-Plasma-Beschleunigern: Von medizinischer Bildgebung bis Hochenergiephysik

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://www.stine.uni-hamburg.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=CO...

DOCTORAL THESES

From 2010 onwards, a total of 21 doctoral theses have been published in the field of accelerator research at Uni Hamburg. Of these, 21 led to science degrees and none to engineering degrees. 4 were written by women and 17 by men.

01. Christoph Maag (2018)

Influence of Pulsed Heat Load on the Performance of Diamond Bragg Mirrors for an XFEL-Oscillator

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
fulltext link	https://ediss.sub.uni-hamburg.de/bitstream/ediss/7793/1/Dissertation.pdf
about facility	European XFEL
in affiliation with	

02. Oliver Stein (2017)

Ultra fast beam loss mechanisms at the LHC and their detection for use in machine protection

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://cds.cern.ch/record/2644713/files/CERN-THESIS-2017-430.pdf?version=1
about facility	LHC
in affiliation with	CERN

03. Theresa Brümmer (2017)

Design study of a laser-driven x-ray source for medical fluorescence imaging

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.3204/PUBDB-2018-01857
fulltext link	https://bib-pubdb1.desy.de/record/402600/files/PhDThesis_BruemmerTheresa.pdf?version=1
about facility	
in affiliation with	

04. Niels Matthias Delbos (2017)

High repetition rate laser-plasma accelerator: 5 hz electron beam generation and advanced target design

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
about facility	
in affiliation with	

05. Spencer Windhorst Jolly (2017)

Spectral phase manipulation of optical pump pulses for mJ-Level narrowband terahertz generation in PPLN

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://ediss.sub.uni-hamburg.de/bitstream/ediss/7518/1/Dissertation.pdf
about facility	
in affiliation with	

06. Irene Dornmair (2017)

Advanced beam dynamics and diagnostics concepts for laser-plasma accelerators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2017
Language	en
doi	10.3204/PUBDB-2017-02493
fulltext link	https://bib-pubdb1.desy.de/record/321223/files/thesis_dornmair_publication.pdf?version=1
about facility	
in affiliation with	

07. Benno Zeitler (2017)

Phase space linearization and external injection of electron bunches into laser-driven plasma wakefields at REGAE

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.3204/PUBDB-2017-00801
fulltext link	https://bib-pubdb1.desy.de/record/317605/files/Dissertation.pdf?version=1
about facility	
in affiliation with	

08. Andy Sven Langner (2016)

A Novel Method and Error Analysis for Beam Optics Measurements and Corrections at the Large Hadron Collider

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	https://cds.cern.ch/record/2265177/files/CERN-THESIS-2016-299.pdf?version=1
about facility	LHC
in affiliation with	

09. Sven Ackermann (2016)

Experimental and Numerical Investigations on Seeding Options for the Multi-Beam Line Free-Electron Laser Facility Flash

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	https://ediss.sub.uni-hamburg.de/bitstream/ediss/7032/1/Dissertation.pdf
about facility	FLASH
in affiliation with	DESY

10. Maria Kuhn (2016)

Transverse emittance measurement and preservation at the LHC

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	http://cds.cern.ch/record/2644713/files/CERN-THESIS-2017-430.pdf?version=1
about facility	LHC
in affiliation with	CERN

11. Roxana Tarkeshian (2012)

Femtosecond resolved diagnostics for electron beam and XUV seed temporal overlap at sFLASH

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2012
Language	en
fulltext link	https://bib-pubdb1.desy.de/record/94541/files/Tarkeshian-Phd.pdf?version=5
about facility	FLASH
in affiliation with	DESY

12. Laurens-Georg Wißmann (2012)

A robust fibre laser system for electro-optic electron bunch profile measurements at FLASH

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	en
fulltext link	https://bib-pubdb1.desy.de/record/140140/files/desy-thesis-12-023.pdf?version=1
about facility	FLASH
in affiliation with	DESY

13. Marie Kristin Bock (2012)

Measuring the electron bunch timing with femtosecond resolution at FLASH

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2012
Language	en
doi	10.3204/DESY-THESIS-2013-008
fulltext link	https://bib-pubdb1.desy.de/record/139310/files/desy-thesis-13-008.pdf?version=3
about facility	FLASH
in affiliation with	DESY

14. Arik Willner (2012)

A high repetition rate XUV seeding source for FLASH2

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	en
fulltext link	https://bib-pubdb1.desy.de/record/96047/files/PhD-AWillner_pub.pdf?version=4
about facility	FLASH
in affiliation with	DESY

15. Christopher Behrens (2012)

Characterization and control of femtosecond electron and x-ray beams at free-electron lasers

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	en
doi	10.3204/DESY-THESIS-2012-038
fulltext link	https://bib-pubdb1.desy.de/record/139966/files/desy-thesis-12-038.pdf?version=1
about facility	
in affiliation with	

16. Yevgeniy Ivanisenko (2012)

Investigation of slice emittance using an energy-chirped electron beam in a dispersive section for photo injector characterization at PITZ

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	en
fulltext link	https://bib-pubdb1.desy.de/record/139054/files/desy-thesis.Ivanisenk.pdf?version=2
about facility	
in affiliation with	

17. Stephan Wesch (2012)

Echtzeitbestimmung longitudinaler Elektronenstrahlparameter mittels absoluter Intensitäts- und Spektralmessung einzelner kohärenter THz Strahlungspulse

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2012
Language	de
doi	10.3204/PUBDB-2017-00801
fulltext link	https://bib-pubdb1.desy.de/record/317605/files/Dissertation.pdf?version=1
about facility	
in affiliation with	

18. Jörn Bödewadt (2011)

Transverse beam diagnostics for the XUV seeding experiment at FLASH

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	en
doi	10.3204/DESY-THESIS-2011-043
fulltext link	https://bib-pubdb1.desy.de/record/96487/files/11_12_12%20-%20PhDThesis_Joern_Boedewadt_FINAL.pdf?version=3
about facility	FLASH
in affiliation with	DESY

19. Marc Hänel (2010)

Experimental investigations on the influence of the photocathode laser pulse parameters on the electron bunch quality in an RF - photoelectron source

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2010
Language	en
doi	10.3204/DESY-THESIS-2010-027
about facility	
in affiliation with	

20. Kirsten Hacker (2010)

Measuring the electron beam energy in a magnetic bunch compressor

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2010
Language	en
doi	10.3204/DESY-THESIS-2010-037
fulltext link	https://bib-pubdb1.desy.de/record/92282/files/kirsten_hacker_diss.pdf?version=3
about facility	
in affiliation with	

21. Andranik Tsakanian (2010)

Time domain numerical calculations of the short electron bunch wakefields in resistive structures

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2010
Language	en
doi	10.3204/DESY-THESIS-2010-041
fulltext link	https://bib-pubdb1.desy.de/record/288935/files/desy-thesis-10-041.pdf?version=1
about facility	
in affiliation with	

GOETHE UNIVERSITY FRANKFURT

GENERAL INFORMATION

Short label for the unit	en	Uni Frankfurt
	de	Uni Frankfurt
Official name of the organization	en	Goethe University Frankfurt
	de	Goethe-Universität Frankfurt am Main
One-line summary	en	Focuses on ion accelerator research; operates UNILAC, experimental and test beam facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Goethe University Frankfurt		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

01. FRANZ		
Type	en	
	de	
current life cycle	en	2.2 Construction
label	en	FRANZ
	de	FRANZ
tagline	en	Frankfurt Neutron Source
	de	Neutronenquelle Frankfurt
description	en	RFQ-IH-based high-current proton accelerator to produce a very intense neutron beam. FRANZ is a proton-beam-driven neutron source at the University of Frankfurt. It accelerates protons at repetition rates up to 250 kHz (100 ns pulses), with a beam energy of around 2 MeV and a beam current of 30 mA. First neutrons from FRANZ are expected in July 2025.
	de	FRANZ ist eine protonenstrahlgetriebene Neutronenquelle an der Universität Frankfurt. Sie beschleunigt Protonen mit Wiederholraten von bis zu 250 kHz (100-ns-Pulse), mit einer Strahlenergie von etwa 2 MeV und einem Strahlstrom von 30 mA. Die ersten Neutronen aus FRANZ werden im Juli 2025 erwartet.

TEACHING

TEACHING OFFERS

In academic year 2024/25, Uni Frankfurt offered 13 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 22.

01. Aktuelle Probleme der Beschleuniger-, Astro- und Plasmaphysik

Teaching event type	Block Course
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

02. Beschleuniger-Seminar

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

03. Einführung in die Beschleunigerphysik mit Exkursion

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

04. Einführung in die Beschleunigerphysik, Übungen zur Vorlesung

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

05. High Intensity Accelerators and their Applications

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

06. High Intensity Accelerators and their Applications, Übungen zur Vorlesung

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

07. Hochleistungsbeschleuniger und deren Anwendungen

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

08. Ringbeschleuniger und Speicherringe

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

09. Linearbeschleuniger

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

10. Laseranwendungen in Teilchenbeschleunigern

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

11. Ringbeschleuniger und Speicherringe, Übungen zur Vorlesung

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-2
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

12. Linearbeschleuniger, Übungen zur Vorlesung

Teaching event type	Lecture Exercise
Weekly semester hours	1
Languages	German
Semesters	2024-2
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

13. Vakuumphysik 2: Vakuumphysik an Beschleunigern

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://qis.server.uni-frankfurt.de/qisserver/rds?state=verpublish&status=init&vmfile=no&pu...

DOCTORAL THESES

From 2010 onwards, a total of 20 doctoral theses have been published in the field of accelerator research at Uni Frankfurt. Of these, 20 led to science degrees and none to engineering degrees. 2 were written by women and 18 by men.

01. Oleksii Lubynets (2024)

Collective phenomena in (multi)strange-hadron production at high μ_B : performance of the CBM experiment at FAIR

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.21248/gups.85576
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/85576/PhD_thesis_Lubynets.pdf
about facility	LHC
in affiliation with	CERNFAIR

02. Dominik Vilsmeier (2024)

Efficient modeling and mitigation of quadrupole errors in synchrotrons and their beam transfer lines

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.21248/gups.85890
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/85890/Dissertation_DominikVilsmeier.pdf
about facility	SIS18
in affiliation with	FAIR

03. Patrick Müller (2024)

Optimization of superconducting crossbar H-mode cavity preparation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.21248/gups.86407
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/86407/Dissertation_Patrick_Mueller_inklLizenz.pdf
about facility	
in affiliation with	

04. Sajjad Hussain Mirza (2021)

Closed orbit feedback system for the fast ramping hadron synchrotrons

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.21248/gups.61016
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/61016/Thesis_PhD_Sajjad_H._Mirza.pdf
about facility	SIS18
in affiliation with	FAIR

05. Heiko Niebuhr (2018)

Entwicklung und Erforschung eines neuartigen Injektionssystems für einen toroidalen Hochstromspeicherring

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	de
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/48967/Dissertation_Heiko_Niebuhr.pdf
about facility	
in affiliation with	

06. Frederik Kesting (2017)

Numerical noise in particle-in-cell tracking: generation and propagation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
about facility	SIS100
in affiliation with	FAIR

07. Maximilian Schütt (2017)

Development of a 325 MHz ladder RFQ

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/45748/Dissertation_schuett.pdf
about facility	p-Linac
in affiliation with	FAIR

08. Hendrik Hähnel (2017)

Development of an IH-type linac for the acceleration of high current heavy ion beams

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/43983/Diss_Haehnel_2017_online.pdf
about facility	UNILAC
in affiliation with	FAIR

09. Florian Burkart (2016)

Expected Damage to Accelerator Equipment Due to the Impact of the Full LHC Beam

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/42034/Doktorarbeit_FBurkart.pdf
about facility	LHC
in affiliation with	CERN

10. Suha Alzubaidi (2016)

Control System for the FRANZ Facility

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2016
Language	en
about facility	FRANZ
in affiliation with	

11. Manuel Heilmann (2016)

Kopplung von 4-Rod-RFQ und IH-DTL für das FRANZ-Projekt

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	de
about facility	
in affiliation with	

12. Batu Klump (2016)

Untersuchungen an niederenergetischen, intensiven und hochbrillanten Protonenstrahlen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	de
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/41358/dissertation_klump_bibliothek.pdf
about facility	
in affiliation with	

13. Daniel Noll (2016)

Investigation on the transport of high-intensity beams using particle-in-cell simulations

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	en
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/42407/diss.pdf
about facility	
in affiliation with	

14. Robert Brodhage (2015)

Entwicklung und Aufbau der ersten gekoppelten CH-Struktur für den neuen Protoneninjektor an der Facility for Antiproton and Ion Research

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
about facility	FAIR
in affiliation with	FAIR

15. Waldemar Schweizer (2015)

Entwicklung einer effizienten Hochstrom-Ionenquelle für das FRANZ-Projekt zur Produktion intensiver, hochbrillanter Protonenstrahlen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2015
Language	de
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/38111/Schweizer.pdf
about facility	
in affiliation with	

16. Vera Chetvertkova (2013)

Verification of Monte Carlo transport codes by activation experiments

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2013
Language	en
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/27848/thesis-Chetvertkova.pdf
about facility	
in affiliation with	FAIR

17. Eike Feldmeier (2013)

Feldkorrekturregelung für dynamische Prozesse in normalleitenden Magneten

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
fulltext link	http://proxy.ub.uni-frankfurt.de/login?url=https%3A%2F%2Fbn-resolving.org%2Furn%3Anbn%3Ade%3Ahebis%3A30%3A3-327984
about facility	
in affiliation with	

18. Christian Schömers (2013)

Entwicklung einer dynamischen Intensitätsregelung für das Heidelberger Ionenstrahl-Therapiesynchrotron

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2013
Language	de
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/32653/DissertationSchoemersElektronisch.pdf
about facility	
in affiliation with	

19. Markus Vossberg (2011)

Der neue RFQ für den Hochladungsinjektor der GSI

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	de
fulltext link	https://publikationen.ub.uni-frankfurt.de/files/22146/Dissertation_Vossberg.pdf
about facility	
in affiliation with	

20. Wafa Daqa (2011)

Linear Betatron coupling studies with space charge effects in the SIS18

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2011
Language	en
about facility	
in affiliation with	

UNIVERSITY OF GÖTTINGEN

GENERAL INFORMATION

Short label for the unit	en	Uni Göttingen
	de	Uni Göttingen
Official name of the organization	en	University of Göttingen
	de	Georg-August-Universität Göttingen
One-line summary	en	Researches accelerator physics; access to laser/plasma labs and local beam facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Göttingen	-	-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

JOHANNES GUTENBERG UNIVERSITY MAINZ

GENERAL INFORMATION

Short label for the unit	en	Uni Mainz
	de	Uni Mainz
Official name of the organization	en	Johannes Gutenberg University Mainz
	de	Johannes Gutenberg-Universität Mainz
One-line summary	en	Pioneers accelerator concepts; operates MAMI, TRIGA reactor, MESA (under construction).
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Johannes Gutenberg University Mainz		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

01. MAMI														
Type	en	Microtron												
	de	Mikrotron												
current life cycle	en	3.1 Operation												
label	en	MAMI												
	de	MAMI												
tagline	en	Mainzer Mikrotron												
	de	Mainzer Mikrotron												
description	en													
	de													

02. MESA

Type	en	Energy Recovery Linac
	de	Linearbeschleuniger mit Energierückgewinnung
current life cycle	en	2.2 Construction
label	en	MESA
	de	MESA
tagline	en	Mainz Energy-Recovering Superconducting Accelerator
	de	Mainz Energy-Recovering Superconducting Accelerator
description	en	
	de	

TEST FACILITIES

TEACHING

TEACHING OFFERS

In academic year 2024/25, Uni Mainz offered 3 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 8.

01. Beschleunigerphysik

Teaching event type	Lecture
Weekly semester hours	4
Languages	German
Semesters	2024-1
Link	https://joqustine.uni-mainz.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=COURSE...

02. Medizin- und Strahlenphysik

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-2

03. Spezielle Probleme der Beschleunigerphysik

Teaching event type	Undergraduate Seminar
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://jogustine.uni-mainz.de/scripts/mgrqispi.dll?APPNAME=CampusNet&PRGNAME=COURSE...

DOCTORAL THESES

From 2010 onwards, a total of 15 doctoral theses have been published in the field of accelerator research at Uni Mainz. Of these, 15 led to science degrees and none to engineering degrees. 1 were written by women and 14 by men.

01. Julian A. List (2024)

Ertüchtigung von supraleitenden crossbar H-mode Hochfrequenzkavitäten mit hohem Kopplungsfaktor

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.25358/openscience-10366
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/10384/1/ertuechtigung_von_supraleitend-20240510114045718.pdf
about facility	UNILAC
in affiliation with	FAIR

02. Tino Morgenroth (2024)

S-EBIT commissioning

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.22032/DBT.64495
about facility	
in affiliation with	

03. Mirco Christmann (2023)

Design studies for the beam-dump experiment DarkMESA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2023
Language	en
doi	10.25358/openscience-9076
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/9093/1/design_studies_for_the_beamdu-20230509115852842.pdf
about facility	
in affiliation with	

04. Nahid Scahill (2022)

Self-modulation of photoemitted bunches at the picosecond timescale

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2022
Language	en
doi	10.25358/OPENSOURCE-6821
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/6832/1/selfmodulation_of_photoemitted-20220315180831747.pdf
about facility	
in affiliation with	

05. Daniel Simon (2021)

Gesamtkonzept für den MESA-Teilchenbeschleuniger unter besonderer Berücksichtigung von Strahloptik und Kryogenik

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	de
doi	10.25358/OPENSOURCE-5809
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/5818/1/simon_daniel-gesamtkonzept_-20210422115707438.pdf
about facility	
in affiliation with	

06. Ben Ledroit (2021)

Target induced halo formation and collimation following MAGIX at MESA

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.25358/openscience-5808
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/5817/1/ledroit_ben-target_induced-20210422114847950.pdf
about facility	MESA
in affiliation with	

07. Christoph Matejcek (2021)

Strahldynamik der Niederenergie-Strahlführung von MESA unter Berücksichtigung von Raumladung und Multipol-Beiträgen höherer Ordnung

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	de
doi	10.25358/openscience-5654
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/5658/1/matejcek_christoph-strahldynamik_-20210218135741640.pdf
about facility	MESA
in affiliation with	

08. Philipp Heil (2021)

Kohärente Smith-Purcell-Strahlung zur minimal invasiven Bunchlängenmessung Im Subpikosekundenbereich

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	de
doi	10.25358/openscience-5886
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/5895/1/heil_philipp-kohärente_smit-20210504093102072.pdf
about facility	MESA
in affiliation with	

09. Timo U. Stengler (2020)

Entwicklung eines supraleitenden Beschleunigermoduls für den rezirkulierenden Betrieb am Mainz Energy-Recovering Superconducting Accelerator (MESA)

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
doi	20.500.12030/4917
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/4917/3/stengler_timo_udo-entwicklung-ei-20200721085410321.pdf
about facility	MESA
in affiliation with	

10. Christian Philipp Stoll (2020)

Beam dynamical behaviour of the MESA SRF structures under recirculating operation

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2020
Language	en
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/5189/3/stoll_christian_philipp-beam_dynamical-20201026154738111.pdf
about facility	MESA
in affiliation with	

11. Tobias Weilbach (2019)

Minimal-invasive strahldiagnose für hoch intensive Elektronenstrahlen

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	en
doi	10.25358/openscience-960
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/962/1/100002941.pdf
about facility	MESA
in affiliation with	FAIR

12. Victor Bechthold (2019)

Untersuchung von Multi-Alkali-Verbindungen im Hinblick auf ihre Eignung zur Erzeugung hochbrillanter Elektronenpulse

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2019
Language	de
doi	20.500.12030/4829
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/4829/1/100002750.pdf
about facility	
in affiliation with	

13. Monika Dehn (2019)

Transportprozesse in Photokathoden und ihr Beitrag zum longitudinalen Halo von Elektronenbeschleunigern

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2019
Language	de
doi	20.500.12030/1204
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/1204/1/dehn_monika-transportproze-20200915105051401.pdf
about facility	
in affiliation with	

14. Igor Constantin Alexander (2018)

Experimental Investigation of the Beam Dynamics of the MESA Photoinjector

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	20.500.12030/1178
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/1178/1/100002353.pdf
about facility	MESA
in affiliation with	

15. Max-Wilhelm Bruker (2016)

Untersuchung der Rückgewinnungseffizienz eines Kühlerelektronenstrahls in longitudinale Magnetfeld

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2016
Language	de
doi	/20.500.12030/4579
fulltext link	https://openscience.ub.uni-mainz.de/bitstream/20.500.12030/4579/1/100000835.pdf
about facility	COSYFAIR
in affiliation with	FAIRHIM

FRIEDRICH SCHILLER UNIVERSITY JENA

GENERAL INFORMATION

Short label for the unit	en	Uni Jena
	de	Uni Jena
Official name of the organization	en	Friedrich Schiller University Jena
	de	Friedrich-Schiller-Universität Jena
One-line summary	en	Focuses on laser-driven accelerators; operates advanced laser plasma facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
Friedrich Schiller University Jena		1	-	-	5	-	-	6	-	-	-	6	-	6
Group: Atomic Physics with highly charged Ions and X-Rays		-	-	-	-	-	-	-	-	-	-	-	-	0
Group: Relativistic laser physics		1	-	-	5	-	-	6	-	-	-	6	-	6

FACILITIES

USER FACILITIES

TEST FACILITIES

01. JETI		
Type	en	Laser Facility
	de	Laseranlage
current life cycle	en	3.1 Operation
label	en	JETI
	de	JETI
tagline	en	Jena Titanium:sapphire 40 TW Laser
	de	Jena Titanium:sapphire 40 TW Laser
description	en	
	de	

02. Ion Beam Centre University of Jena

Type	en	Ion Beam Facility
	de	Ionenstrahl-Anlage
current life cycle	en	
label	en	Ion Beam Centre University of Jena
	de	Ionenstrahlzentrum Uni Jena
tagline	en	
	de	
description	en	
	de	

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

From 2010 onwards, a total of 9 doctoral theses have been published in the field of accelerator research at Uni Jena. Of these, 9 led to science degrees and none to engineering degrees. 5 were written by women and 4 by men.

01. Mohammed Almassarani (2024)

Sub-Picosecond Dynamics During Relativistic Laser-Plasma Interaction

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2024
Language	en
doi	10.22032/DBT.61635
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00064049/Dissertation_MohammedAlma
about facility	
in affiliation with	

02. Supriya Rajhans (2024)

Few-cycle laser pulses for compact plasma accelerators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2024
Language	en
doi	10.22032/DBT.63574
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00065736/dissertationRajhans.pdf
about facility	
in affiliation with	

03. Harsh Harsh (2024)

Bright and stable betatron beams from laser-driven plasma wakefield

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2024
Language	en
doi	10.22032/dbt.64120
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00066211/Dissertation_Harsh.pdf
about facility	
in affiliation with	

04. Georg Becker (2021)

Characterization of laser-driven proton acceleration with contrast-enhanced laser pulses

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2021
Language	en
doi	10.22032/dbt.49350
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00053896/dissbecker.pdf
about facility	
in affiliation with	

05. Issa Tamer (2020)

Petawatt-class laser optimization and ultrashort probe pulse generation for relativistic laser-plasma interactions

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2020
Language	en
doi	10.22032/dbt.40833
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00046690/disstamer.pdf
about facility	
in affiliation with	

06. Maria Reuter (2018)

Characterisation of a laser wakefield accelerator with ultra-short probe pulses

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2018
Language	en
doi	10.22032/DBT.38073
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00044194/Diss_reuter.pdf
about facility	
in affiliation with	

07. Sven Breilkopf (2018)

Energy scaling by stacking of ultrashort pulses in a passive enhancement cavity

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2018
Language	en
doi	10.22032/DBT.34968
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00041164/dissbreilkopf.pdf
about facility	
in affiliation with	

08. Stephan Kuschel (2018)

Erzeugung dichter elektronenpulse mit laser-plasma-beschleunigern für QED experimente in hohen feldern

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	female
Year	2018
Language	en
doi	10.22032/DBT.34353
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00040537/Dissertation_Kuschel.pdf
about facility	
in affiliation with	

09. László Veisz (2017)

Investigation of parametric instabilities in femtosecond laser-produced plasmas

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr. rer. nat.)
Author's gender	male
Year	2017
Language	en
doi	10.22032/DBT.33656
fulltext link	https://www.db-thueringen.de/servlets/MCRFileNodeServlet/dbt_derivate_00039813/Veisz_Aa.pdf
about facility	
in affiliation with	

UNIVERSITY OF ROSTOCK

GENERAL INFORMATION

Short label for the unit	en	Uni Rostock
	de	Uni Rostock
Official name of the organization	en	University of Rostock
	de	Universität Rostock
One-line summary	en	Researches ion and plasma accelerators; operates beam labs for interdisciplinary applications.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Rostock		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

From 2010 onwards, a total of 14 doctoral theses have been published in the field of accelerator research at Uni Rostock. Of these, 0 led to science degrees and 14 to engineering degrees. 4 were written by women and 10 by men.

01. Gowrishankar Thalagavadi Hallilingaiah (2022)

Investigation and development of a transverse deflecting structure

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2022
Language	en
doi	10.18453/rosdok_id00004375
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000003009/rosdok_derivate_0000209092/Hallilingaiah_Disserta
about facility	ANKAELBE
in affiliation with	

02. Shahnam Gorgi Zadeh (2020)

Accelerating cavity and higher order mode coupler design for the future circular collider

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2020
Language	en
doi	10.18453/rosdok_id00003023
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002516/rosdok_derivate_0000100582/GorgiZadeh_Dissertati
about facility	
in affiliation with	

03. Kai Papke (2020)

Methods for the design and analysis of higher-order mode couplers applied to superconducting accelerating structures

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2020
Language	en
doi	10.18453/rosdok_id00002899
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002434/rosdok_derivate_0000096616/Papke_Dissertation_20
about facility	
in affiliation with	

04. Hermann Winrich Pommerenke (2020)

Compact radio-frequency quadrupoles for industrial and medical applications

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2020
Language	en
doi	10.18453/rosdok_id00002956
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002457/rosdok_derivate_0000099121/Pommerenke_Dissertat
about facility	
in affiliation with	

05. Eden Tafa Tulu (2020)

Developing an electron multipacting-free cathode unit of the superconducting radio frequency photoinjector

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2020
Language	en
doi	10.18453/rosdok_id00002903
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002436/rosdok_derivate_0000096895/Tulu_Dissertation_2020
about facility	
in affiliation with	

06. Johann Heller (2018)

Numerical simulation of electromagnetic fields in complex multi-cavity superconducting radio frequency resonators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
doi	10.18453/rosdok_id00002402
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002055/rosdok_derivate_0000065963/Heller_Dissertation_2018
about facility	
in affiliation with	

07. Franziska Reimann (2018)

Dielectrically lined rectangular waveguides as wakefield based dechirper structures

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2018
Language	en
doi	10.18453/rosdok_id00002394
fulltext link	https://doi.org/10.18453/rosdok_id00002394
about facility	
in affiliation with	

08. Mario Beck (2018)

Numerical and experimental studies to model and reduce the impedance in the CERN Super Proton Synchrotron (SPS)

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
doi	10.18453/rosdok_id00002494
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000002137/rosdok_derivate_0000068296/Beck_Dissertation_2018.pdf
about facility	SPS
in affiliation with	

09. Korinna Brackebusch (2018)

Perturbative methods for the computation of resonant cavity eigenmodes subject to geometric variations

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2018
Language	en
doi	10.18453/rosdok_id00002062
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000001884/rosdok_derivate_0000038652/Dissertation_Brackebusch.pdf
about facility	
in affiliation with	

10. Dawei Zheng (2018)

Efficient algorithms for the fast computation of space charge effects caused by charged particles in particle accelerators

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2018
Language	en
doi	10.18453/rosdok_id00002019
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000001841/rosdok_derivate_0000038447/Dissertation_Zheng_2018.pdf
about facility	
in affiliation with	

11. Thomas Flisgen (2015)

Compact state-space models for complex superconducting radio-frequency structures based on model order reduction and concatenation methods

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2015
Language	en
fulltext link	https://cds.cern.ch/record/2104467/files/CERN-ACC-2015-0145.pdf
about facility	
in affiliation with	

12. Aleksandar Markovik (2014)

Simulation of the interaction of positively charged beams and electron clouds

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2014
Language	en
doi	10.18453/rosdok_id00001279
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000001100/rosdok_derivate_0000005257/Dissertation_Markovik_2014.pdf
about facility	
in affiliation with	

13. André Arnold (2012)

Simulation und Messung der Hochfrequenzeigenschaften einer supraleitenden Photo-Elektronenquelle

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	male
Year	2012
Language	de
doi	10.18453/rosdok_id00001087
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000000908/rosdok_derivate_0000004968/Dissertation_Arnold_2
about facility	
in affiliation with	

14. Jenniffa Rudolph (2012)

Instruments and techniques for analysing the time-resolved transverse phase space distribution of high-brightness electron beams

Class	Doctoral thesis
Subclasses	Doctoral thesis (Dr.-Ing.)
Author's gender	female
Year	2012
Language	en
doi	10.18453/rosdok_id00001106
fulltext link	https://rosdok.uni-rostock.de/file/rosdok_disshab_0000000927/rosdok_derivate_0000004994/Dissertation_Rudolph_
about facility	
in affiliation with	

UNIVERSITY OF KASSEL

GENERAL INFORMATION

Short label for the unit	en	Uni Kassel
	de	Uni Kassel
Official name of the organization	en	University of Kassel
	de	Universität Kassel
One-line summary	en	Engaged in accelerator-related physics; operates local electron and ion beam labs.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Kassel		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

UNIVERSITY OF WUPPERTAL

GENERAL INFORMATION

Short label for the unit	en	Uni Wuppertal
	de	Uni Wuppertal
Official name of the organization	en	University of Wuppertal
	de	Bergische Universität Wuppertal
One-line summary	en	Achieves accelerator R&D in particle, ion, and laser acceleration; operates teaching and research facilities.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Wuppertal	-	-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

Currently, only courses offered by universities are listed. However, the data model provides for the inclusion of affiliated organizations and summer schools, so that corresponding entries could be displayed here.

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

UNIVERSITY OF SIEGEN

GENERAL INFORMATION

Short label for the unit	en	Uni Siegen
	de	Uni Siegen
Official name of the organization	en	University of Siegen
	de	Universität Siegen
One-line summary	en	Participates in applied accelerator research; hosts electron and ion beam facilities for study.
	de	<to be translated>

RESEARCH GROUPS

NAME	MEMBERS	PROFESSOR	SENIOR RESEARCHER	POST DOC	PHD STUDENT	MASTER STUDENT	BACHELOR STUDENT	PHYSICIST	ENGINEER	OTHER	FEMALE	MALE	NONBINARY	RECURSIVE SUM
University of Siegen		-	-	-	-	-	-	-	-	-	-	-	-	0

FACILITIES

USER FACILITIES

TEST FACILITIES

TEACHING

TEACHING OFFERS

In academic year 2024/25, Uni Siegen offered 3 courses related to accelerator research of which all addresses bachelor students. and all addresses master students.. The combined weekly semester hours was 6.

01. Accelerator Physics I

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://unisono.uni-siegen.de:443/qisserver/pages/startFlow.xhtml?_flowId=detailView-flow&...

02. Accelerator Physics I (Exercise)

Teaching event type	Lecture Exercise
Weekly semester hours	2
Languages	German
Semesters	2024-2
Link	https://unisono.uni-siegen.de/qisserver/pages/startFlow.xhtml?_flowId=detailView-flow&_flo...

03. Introduction to Accelerator Physics II

Teaching event type	Lecture
Weekly semester hours	2
Languages	German
Semesters	2024-1
Link	https://unisono.uni-siegen.de:443/qisserver/pages/startFlow.xhtml?_flowId=detailView-flow&...

DOCTORAL THESES

Doctoral theses are currently displayed for universities only. However, the data model provides for the specification of organizations involved in doctoral studies, so that entries marked accordingly could be displayed here.

Built at: 2025-09-09T12:16:45.685Z